

# Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons

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in

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by

**Soumen Kumar Mondal**

(Roll No. 23m2157)

Under the Guidance of

**Prof. Preethi Jyothi**

(Associate Professor, CSE, IIT Bombay)



Center for Machine Intelligence and Data Science (CMInDS)

**INDIAN INSTITUTE OF TECHNOLOGY BOMBAY**

**Mumbai - 400076, India**

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# *Approval*

This BPS (Autumn, 2024) entitled "**Improving Downstream Task Performance in Multilingual LLMs by Intervening Language Specific Neurons**" by **Soumen Kumar Mondal**, Roll No. 23m2157, is approved for the degree of *Master of Science by Research in Data Science and Artificial Intelligence*.

.....  
Prof. Preethi Jyothi  
Associate Professor, CSE, IIT Bombay  
(Supervisor)

.....  
Prof. Pushpak Bhattacharyya  
Professor, CSE, IIT Bombay  
(Internal Examiner)

Date: December 26, 2024  
Place: Mumbai, India

# *Declaration*

I declare that this written submission represents my ideas in my own words. Where others' ideas and words have been included, I have adequately cited and referenced the original source. I declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated, or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the source which has thus not been properly cited or from whom proper permission has not been taken when needed.

**Soumen Kumar Mondal**

Date: December 26, 2024

Place: Mumbai, India

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**Soumen Kumar Mondal**

Date: December 27, 2024

Place: Mumbai, India

# *Abstract*

This thesis investigates methodologies for identifying and utilizing language-specific neurons in multilingual large language models (LLMs) to improve performance on multilingual downstream tasks, particularly for low-resource languages. The research examines how language-specific information is represented in various LLM architectures, including Llama3.1, Bloomz, and Aya models. Techniques such as activation probability entropy and LoRA-based fine-tuning methods were applied to identify language-specific neurons using datasets like Wikipedia and XNLI. Intervention strategies, such as replacing neuron activations with language-specific statistical activations, were explored to determine their effect on task performance. Experiments involving neuron-specific fine-tuning and neuron freezing were also conducted to study the relationship between architectural modifications and language representation.

The study further analyzes perplexity distributions and neuron activation maps across different languages and models, providing insights into how multilingual datasets influence the internal mechanisms of LLMs. Experimental setups were developed to assess the effectiveness of fine-tuning and intervention methods for improving natural language inference (NLI) performance in low-resource languages without requiring direct training data. By addressing existing gaps in language-specific neuron identification, this work contributes to a better understanding of how LLMs can be tailored to achieve performance improvement across diverse languages. The results provide a deeper understanding of the interaction between LLM architecture and multilingual capabilities, particularly for underrepresented languages.

# Contents

Approval

Declaration

Acknowledgements

Abstract

Contents

List of Figures

List of Tables

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                        | <b>1</b>  |
| 1.1      | Introduction . . . . .                                     | 1         |
| 1.2      | Motivation . . . . .                                       | 1         |
| 1.3      | Problem Statement . . . . .                                | 2         |
| <b>2</b> | <b>Literature Review</b>                                   | <b>3</b>  |
| 2.1      | Neurons in Multilingual Models . . . . .                   | 3         |
| 2.2      | Mechanistic Interpretability and Multilinguality . . . . . | 5         |
| <b>3</b> | <b>Methodology</b>                                         | <b>6</b>  |
| 3.1      | Definition of Neuron . . . . .                             | 6         |
| 3.2      | Dataset for Neuron Identification . . . . .                | 8         |
| 3.3      | Statistics of Neuron Activation . . . . .                  | 8         |
| 3.4      | Methods for Neuron Identification . . . . .                | 9         |
| 3.4.1    | LAPE . . . . .                                             | 9         |
| 3.4.2    | Activation Based Methods . . . . .                         | 12        |
| 3.4.3    | Task Neurons for a Language . . . . .                      | 13        |
| 3.5      | Neuron Fine-Tuning using LoRA for a Task . . . . .         | 14        |
| <b>4</b> | <b>Experiments</b>                                         | <b>17</b> |
| 4.1      | Task and Dataset: XNLI . . . . .                           | 17        |
| 4.2      | Baseline Experiments . . . . .                             | 18        |

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| 4.3      | Experiment Goals . . . . .                                            | 18        |
| 4.4      | Targeted Neuron Fine-tuning Experiment . . . . .                      | 20        |
| 4.5      | Targeted Neuron Fine-Tuning Configurations . . . . .                  | 21        |
| <b>5</b> | <b>Analysis and Results</b>                                           | <b>22</b> |
| 5.1      | LAPE Neuron Analysis . . . . .                                        | 22        |
| 5.1.1    | Language Sets . . . . .                                               | 22        |
| 5.1.2    | Language Neuron Distribution . . . . .                                | 23        |
| 5.1.3    | Perplexity Change . . . . .                                           | 25        |
| 5.2      | Activation Probability 90p Neuron Analysis . . . . .                  | 28        |
| 5.2.1    | Language Neuron Distribution . . . . .                                | 28        |
| 5.2.2    | Perplexity Change . . . . .                                           | 28        |
| 5.3      | Fine-Tuning Experiment Results on XNLI Task . . . . .                 | 30        |
| 5.3.1    | Baseline Experiment Results . . . . .                                 | 30        |
| 5.3.2    | LAPE Experiment Results using Lang Specific Neurons . . . . .         | 31        |
| 5.3.3    | Act Prob 90p Experiment Results using Lang Specific Neurons . . . . . | 33        |
| 5.3.4    | Act Prob 90p Experiment Results using Task Specific Neurons . . . . . | 34        |
| <b>6</b> | <b>Conclusions</b>                                                    | <b>36</b> |
| 6.1      | Conclusion and Future Work . . . . .                                  | 36        |
| 6.1.1    | Conclusion . . . . .                                                  | 36        |
| 6.1.2    | Future Work . . . . .                                                 | 37        |
|          | <b>Bibliography</b>                                                   | <b>38</b> |

# List of Figures

|     |                                                                                                            |    |
|-----|------------------------------------------------------------------------------------------------------------|----|
| 3.1 | Architecture of MLP module in LLaMA model . . . . .                                                        | 7  |
| 3.2 | Procedure of computing LAPE for a neuron. . . . .                                                          | 11 |
| 3.3 | LoRA fine tuning method for target neurons. . . . .                                                        | 16 |
| 5.1 | Language-specific neuron distributions for LAPE method for Set1 and Set3. . .                              | 24 |
| 5.2 | Neuron overlap between languages for LAPE method for Set1 and Set3. . . . .                                | 25 |
| 5.3 | Layer-wise neuron distributions for LAPE method for Set1 and Set3. . . . .                                 | 26 |
| 5.4 | Perplexity change $PPXC(i, j)$ for LAPE method for Set1 and Set3. . . . .                                  | 27 |
| 5.5 | Overlap between neurons and layer-wise neurons distribution for activation probability 90p method. . . . . | 29 |
| 5.6 | Perplexity change $PPXC(i, j)$ for activation probability based methods. . . . .                           | 30 |

# List of Tables

|     |                                                                                                                                                                                 |    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5.1 | Baseline results for the XNLI task for Llama 3.1. . . . .                                                                                                                       | 30 |
| 5.2 | XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English. . . . .          | 32 |
| 5.3 | XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English . . . . .          | 33 |
| 5.4 | XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for act_prob_90p method for Llama3.1. The training language is always English . . . . . | 34 |
| 5.5 | XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for act_prob_90p method for Llama3.1. The training language is always English . . . . .  | 35 |
| 5.6 | XNLI task results (intervention by higher quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English . . . . .   | 35 |
| 5.7 | XNLI task results (intervention by lower quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English . . . . .    | 35 |

# Chapter 1

## Introduction

### 1.1 Introduction

The emergence of large language models (LLMs) has transformed natural language processing (NLP), enabling remarkable advancements across a wide range of linguistic tasks. Despite their success, challenges persist in ensuring comparable performance for low-resource languages, which often lack sufficient representation in pretraining datasets. This thesis addresses these challenges by investigating the representation of language-specific information within LLMs and exploring methods to identify and utilize language-specific neurons. By focusing on multilingual tasks such as natural language inference (NLI), this research seeks to better understand how language-specific information is encoded and how it can be effectively utilized to improve performance, particularly for underrepresented languages.

The research builds on recent advancements in neuron analysis and fine-tuning techniques, using models such as Llama3.1, Bloomz, and Aya to explore the relationship between architectural choices, pretraining datasets, and language-specific representations. By employing methods such as activation probability entropy and LoRA-based fine-tuning, the study systematically identifies language-specific neurons and evaluates their role in downstream tasks. Through detailed experiments on layer-specific fine-tuning, freezing, and intervention strategies, the thesis aims to provide a comprehensive understanding of the mechanisms underlying multilingual capabilities in LLMs.

### 1.2 Motivation

In the era of multilingual LLMs, achieving consistent performance across diverse languages remains a significant challenge. High-resource languages benefit from extensive pretraining data and resources, leading to superior performance on downstream tasks, while low-resource

languages often lag behind. Addressing this imbalance is essential for ensuring that the advancements in NLP benefit speakers of all languages, regardless of the resources available for their linguistic representation.

Understanding the internal mechanisms of LLMs, specifically how they represent language-specific information, offers a way to addressing these challenges. By identifying language-specific neurons and their roles in encoding linguistic knowledge, we can develop targeted interventions to improve performance for low-resource languages. This research is driven by the need to close the performance gap, advance the understanding of multilingual model architectures, and contribute to the development of fair and inclusive NLP systems.

### 1.3 Problem Statement

While LLMs exhibit strong multilingual capabilities, their performance across languages is often inconsistent, with low-resource languages showing suboptimal results compared to high-resource counterparts. This disparity can be attributed to the lack of sufficient representation for low-resource languages during pretraining and the challenge of effectively isolating and utilizing language-specific information within these models. Existing methods for identifying language-specific neurons often rely on arbitrary thresholds or lack generalizability across different architectures and datasets. Moreover, the impact of architectural modifications, such as layer-specific fine-tuning or neuron freezing, on language-specific representation and downstream performance remains underexplored.

This research aims to address these limitations by developing systematic approaches to identify and analyze language-specific neurons in LLMs. By focusing on techniques such as neuron activation probability and LoRA-based fine-tuning, the study seeks to bridge the gap in understanding how language-specific information is encoded. The ultimate goal is to improve task performance for low-resource languages without requiring extensive annotated data, thereby contributing to the application of LLMs in multilingual settings.

# Chapter 2

## Literature Review

### 2.1 Neurons in Multilingual Models

Understanding the representation of language-specific and cross-lingual knowledge in multilingual language models (ML-LMs) is crucial for improving their generalization capabilities across diverse languages. Recent studies, such as those probing mBERT and XLM-R using the mLAMA dataset [Zhao et al. (2024a)], have revealed that factual knowledge in ML-LMs is encoded in three distinct patterns: language-independent, cross-lingual shared, and cross-lingual transferred representations. Language-independent neurons encode facts uniquely for each language, whereas cross-lingual shared neurons represent the same facts using identical neural patterns across languages. Cross-lingual transferred neurons leverage knowledge from high-resource languages to infer facts in low-resource languages. However, limitations in current models hinder consistent fact representation and effective cross-lingual transfer, underscoring the need for targeted interventions and better probing techniques to enhance their multilingual factual capabilities.

The study by Kojima et al. (2024) explored language-specific neurons in decoder-based pre-trained language models (PLMs) such as XGLM, BLOOM, and Llama2. These neurons, which are uniquely activated for specific languages, were found predominantly in the initial and final layers of the models. The researchers demonstrated that manipulating these neurons during inference could significantly influence the language probability of generated text. The findings highlight the intricate role of language-specific neurons in cross-lingual text generation, suggesting that decoder-based PLMs exhibit a layered structure of multilingual capabilities, where early and late layers handle language-specific processing while middle layers remain language-agnostic. This work underscores the potential of targeted neuron interventions to enhance multilingual performance in PLMs.

Task-specific neurons Leng and Xiong (2024) in large language models (LLMs) have been empirically shown to play a critical role in multi-task learning and generalization. Using gradient attribution, these neurons are identified as those with the highest relevance to specific tasks,

and their deactivation or fine-tuning significantly impacts task-specific performance. Importantly, overlaps in task-specific neurons across tasks are strongly correlated with generalization capabilities, providing evidence of shared knowledge representations in LLMs. Additionally, methods like Neuron-level Continuous Fine-Tuning (NCFT) leverage these neurons to mitigate catastrophic forgetting in continuous learning scenarios by isolating and dynamically weighting task-specific parameters during training and inference, as demonstrated in recent studies.

Recent advancements in understanding multilingual capabilities in LLMs emphasize the pivotal role of language-specific neurons. [Tang et al. \(2024\)](#) propose the novel Language Activation Probability Entropy (LAPE) metric to identify neurons that are preferentially activated by specific languages, enabling insights into the organization of multilingual capabilities within models like LLaMA-2 and BLOOM. Their findings reveal that these neurons are predominantly located in the top and bottom layers, facilitating both the transformation of language-specific inputs into shared semantic representations and the generation of outputs in respective languages. Furthermore, selective activation or deactivation of these neurons demonstrates the ability to steer model responses in a particular language, showcasing the potential for targeted interventions to enhance cross-lingual performance.

Studies on LLMs have revealed the presence of language-specific neurons, which play a crucial role in handling multilingual tasks. These neurons exhibit distinct activation patterns when processing inputs in specific languages, as identified through methods such as Parallel Language-Specific Neuron Detection (PLND). The proposed Multilingual Workflow (MWork) hypothesizes that LLMs process queries in three stages: understanding the query by converting it into English, reasoning in English while incorporating multilingual knowledge through feed-forward structures, and finally generating responses in the original query language. Experiments demonstrate that fine-tuning these neurons can significantly enhance multilingual capabilities across both high-resource and low-resource languages without negatively impacting performance in other languages. These findings underscore the importance of language-specific neurons in improving multilingual generalization and task performance [Zhao et al. \(2024b\)](#).

Advancements in analyzing LLMs have identified the presence of style-specific neurons, which extend to multilingual capabilities. These neurons, as highlighted by [Lai et al. \(2024\)](#), are crucial for tasks such as text style transfer (TST) and language adaptation. Style-specific neurons are identified by their selective activations for particular attributes, such as language or stylistic features, and can be manipulated to steer LLM outputs. For instance, deactivating source-specific neurons enhances the generation of target-style words but can impact fluency. Methods like contrastive decoding, adapted for style-specific tasks, mitigate these effects by refining token probabilities across layers. Such neuron-level analysis not only demonstrates the modularity within LLMs but also underscores their adaptability for multilingual and stylistic transformations.

Recent research has revealed the critical role of neurons in the factual knowledge representation and retrieval within multilingual LLMs. For example, [Meng et al. \(2022\)](#) identified specific mid-layer feedforward networks in GPT models that localize factual associations, which can be directly edited without disrupting unrelated facts. Their work proposed Rank-One Model Editing (ROME), a method to modify the memory of such neurons effectively, demonstrating high generalization and specificity for knowledge updates. This study underscores the importance

of mid-layer computations for cross-lingual tasks, enabling more interpretable and controllable model adaptations.

## 2.2 Mechanistic Interpretability and Multilinguality

One notable contribution in the field of mechanistic interpretability is the introduction of Soft-max Linear Units (SoLU), an architectural modification designed to enhance neuron interpretability. By replacing traditional activation functions with SoLU, researchers observed a substantial increase in neurons that correspond to human-understandable concepts, phrases, or categories. This enhancement facilitates a clearer understanding of how models process information, including multilingual data, by making the internal representations more transparent and interpretable [Elhage et al. (2022a)].

Building on this, Anthropic employed dictionary learning techniques to decompose language models into understandable components. By identifying patterns of neuron activations, termed "features," researchers could map these to specific concepts within the model. This approach was successfully scaled to larger models like Claude 3.0 Sonnet, revealing features corresponding to a vast range of entities and concepts across multiple languages. These findings underscore the model's capacity to represent multilingual information through distinct, interpretable features, enhancing our understanding of how LLMs manage multilinguality [Bricken et al. (2023)].

Additionally, Anthropic's exploration into the phenomenon of superposition—where models represent more features than they have dimensions—has implications for multilingual processing. Understanding how models compress and manage multiple linguistic features within limited dimensions can inform strategies to improve multilingual performance and interpretability. By dissecting these mechanisms, researchers can develop more efficient models capable of handling multiple languages without compromising interpretability [Elhage et al. (2022b)].

# Chapter 3

## Methodology

### 3.1 Definition of Neuron

In the context of multi layer perceptrons (MLPs) in large language models (LLMs), a *neuron* refers to a unit of computation within the MLP module. Specifically, consider the  $i^{\text{th}}$  layer of a Transformer architecture, where the input is a hidden state vector  $\mathbf{H}^{i-1} \in \mathbb{R}^{T \times d}$ , derived from the previous layer.

**Hidden State Representation:** The output of the multi-head self-attention (MHA) mechanism serves as input to the MLP and is given as:

$$\hat{\mathbf{H}}^i = \text{Attn}(\mathbf{H}^{i-1}\mathbf{W}_q^i, \mathbf{H}^{i-1}\mathbf{W}_k^i, \mathbf{H}^{i-1}\mathbf{W}_v^i) \cdot \mathbf{W}_o^i, \quad (3.1)$$

where  $\mathbf{W}_q^i, \mathbf{W}_k^i, \mathbf{W}_v^i, \mathbf{W}_o^i \in \mathbb{R}^{d \times d}$  are the trainable parameters of the self-attention module, and  $\text{Attn}(\cdot)$  denotes the scaled dot-product attention operation.

**MLP Transformation:** The MLP module takes  $\hat{\mathbf{H}}^i$  as input and transforms it as follows:

$$\mathbf{h}^i = \text{act\_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \cdot \mathbf{W}_2^i, \quad (3.2)$$

where  $\mathbf{W}_1^i \in \mathbb{R}^{d \times 4d}$  and  $\mathbf{W}_2^i \in \mathbb{R}^{4d \times d}$  are learnable projection matrices, and  $\text{act\_fn}(\cdot)$  represents a non-linear activation function such as GELU or SiLU.

**Neuron:** Within the MLP, the  $j^{\text{th}}$  neuron of the  $i^{\text{th}}$  layer corresponds to the  $j^{\text{th}}$  row of the intermediate activation matrix:

$$\mathbf{h}_j^i = \text{act\_fn}((\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j), \quad (3.3)$$

where  $(\hat{\mathbf{H}}^i \mathbf{W}_1^i)_j$  represents the pre-activation value of the  $j^{\text{th}}$  neuron.

**Generalized Neuron Formulation:** Recent architectures such as LLaMA [Grattafiori and Others (2024)] utilize Gated Linear Units (GLU) to enhance MLP performance, modifying the

MLP computation as:

$$\mathbf{h}^i = [\text{act\_fn}(\hat{\mathbf{H}}^i \mathbf{W}_1^i) \otimes (\hat{\mathbf{H}}^i \mathbf{W}_3^i)] \cdot \mathbf{W}_2^i, \quad (3.4)$$

where  $\mathbf{W}_3^i \in \mathbb{R}^{d \times 4d}$  is an additional projection matrix, and  $\otimes$  denotes element-wise multiplication. This formulation introduces gating to the MLP, allowing for more expressive transformations. Figure 3.1 shows the architecture of the MLP module.

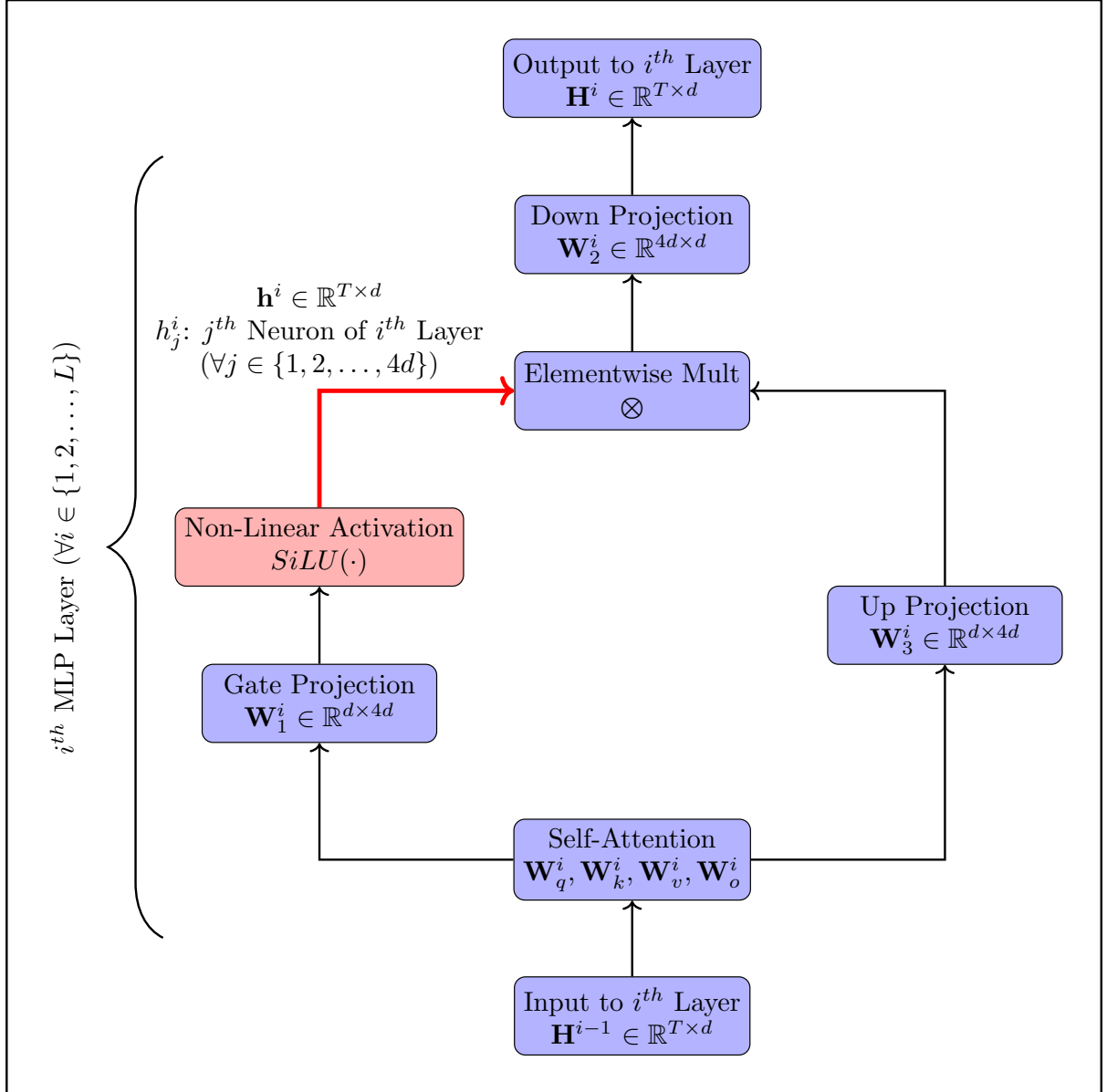


FIGURE 3.1: Architecture of MLP module in LLaMA model

## 3.2 Dataset for Neuron Identification

For the purpose of identifying language-specific neurons, we utilize a subset of the Wikipedia dataset spanning 15 languages:

$$\{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

Each language’s dataset is preprocessed and passed through pretrained large language models (LLMs). The activations are then collected from the MLP modules of the models.

**Dataset Description:** The Wikipedia dataset [Foundation] for each language contains a sequence of tokenized text data. Each tokenized sequence has a maximum length of  $T_{\max} = 512$ . For every sequence, we compute the activations from the MLP module, followed by averaging the activations at each neuron position across all sequences. This provides a language-specific profile for every neuron  $(i, j)$  in the model.

**Activation Collection:** Given a pretrained LLM, let the MLP activation for the  $i^{\text{th}}$  layer and the  $j^{\text{th}}$  neuron in a specific language be represented by  $h_{i,j}^t \in \mathbb{R}$ , where  $t$  indexes the token positions in a sequence of length  $T$ . The average activation of a neuron across all token positions for a given sequence is computed as:

$$\bar{h}_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.5)$$

where  $h_{i,j}^t$  is the neuron activation corresponding to the  $t^{\text{th}}$  token.

**Language-Specific Activations:** For each language  $l$  in the dataset, we compute the average activation of each neuron  $(i, j)$  over all sequences  $S_l$  in the dataset:

$$h_{i,j}^l = \frac{1}{|S_l|} \sum_{s \in S_l} \bar{h}_{i,j}^{(s)}, \quad (3.6)$$

where  $h_{i,j}^l$  represents the average activation of the  $j^{\text{th}}$  neuron in the  $i^{\text{th}}$  layer for language  $l$ , and  $|S_l|$  denotes the number of sequences for language  $l$ .

## 3.3 Statistics of Neuron Activation

The statistics of neuron activation are precomputed to analyze language-specific patterns in the MLP layers of the pretrained large language models (LLMs). These statistics include mean, standard deviation, and quantiles (e.g., 5th, 10th, 25th, 50th, 75th, 90th, and 95th percentiles) of the neuron activations for each language in the dataset. The computations are performed across all layers and neurons within the MLP modules.

**Activation Mean and Standard Deviation:** The mean activation of a neuron  $(i, j)$  in the  $t^{\text{th}}$  token position is computed as:

$$\mu_{i,j} = \frac{1}{T} \sum_{t=1}^T h_{i,j}^t, \quad (3.7)$$

where  $h_{i,j}^t$  is the activation of the  $(i, j)$  neuron for the  $t^{\text{th}}$  token, and  $T$  is the total number of tokens in a sequence. Similarly, the standard deviation of the activations is calculated as:

$$\sigma_{i,j} = \sqrt{\frac{1}{T} \sum_{t=1}^T (h_{i,j}^t - \mu_{i,j})^2}. \quad (3.8)$$

**Quantiles of Activation:** Quantile-based statistics provide insights into the distribution of activations for each neuron. The  $q$ -quantile activation value, denoted as  $P_q$ , is defined as:

$$P_q(h_{i,j}) = \inf \{x \in \mathbb{R} : F_{h_{i,j}}(x) \geq q\}, \quad (3.9)$$

where  $F_{h_{i,j}}(x)$  is the cumulative distribution function (CDF) of the activations for neuron  $(i, j)$ . Commonly computed quantiles include:

$$P_{50}, P_{75}, P_{90}, P_{95}, P_{25}, P_{10}, P_5.$$

**Language-Specific Aggregation:** For a given language  $l$  and all sequences in the dataset, the mean and other statistics are aggregated across all token positions and batches. Let  $S_l$  denote the set of sequences for language  $l$ , and  $B$  represent the batch size. The aggregated mean activation for neuron  $(i, j)$  is computed as:

$$\bar{\mu}_{i,j}^l = \frac{1}{|S_l|} \sum_{b=1}^B \sum_{s \in S_l} \mu_{i,j}^{(s,b)}, \quad (3.10)$$

where  $\mu_{i,j}^{(s,b)}$  is the mean activation of neuron  $(i, j)$  for the  $s^{\text{th}}$  sequence in the  $b^{\text{th}}$  batch.

## 3.4 Methods for Neuron Identification

### 3.4.1 LAPE

Language Activation Probability Entropy (LAPE) [Tang et al. (2024)] is a method designed to identify neurons that exhibit language-specific behavior across multiple languages. This metric quantifies the entropy of normalized activation probabilities for each neuron across the given set of languages.

**Step 1: Compute Activation Probabilities.** For a pretrained large language model with  $L$  layers and  $4d$  neurons per layer, the activation of a neuron  $(i, j)$  in response to a sequence

$X^l \in \mathbb{R}^{N \times T}$  (where  $N$  is the number of sequences, and  $T$  is the number of tokens per sequence) for a language  $l$  is denoted as  $h_{i,j}^l$ . The unnormalized activation probability for neuron  $(i, j)$  is given by:

$$p_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.11)$$

where  $\mathbb{I}$  is the indicator function that returns 1 if the activation  $h_{i,j}^l$  is greater than zero, and 0 otherwise. The expectation is computed over all tokens in all sequences for language  $l$ .

**Step 2: Normalize Activation Probabilities.** The normalized activation probability is computed across  $k$  languages to ensure a valid probability distribution:

$$P_{i,j}^l = \frac{p_{i,j}^l}{\sum_{l=1}^k p_{i,j}^l}, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.12)$$

**Step 3: Compute Language Activation Probability Entropy.** The entropy of the normalized activation probabilities for each neuron  $(i, j)$  is computed as:

$$\text{LAPE}(i, j) = - \sum_{l=1}^k P_{i,j}^l \log P_{i,j}^l, \quad \forall i \in \{1, \dots, L\}, \forall j \in \{1, \dots, 4d\}. \quad (3.13)$$

This measures the uncertainty in the activation distribution of a neuron across the  $k$  languages. Figure 3.2 shows the procedure of computing LAPE for a neuron.

**Step 4: Apply Threshold and Select Neurons.** To identify neurons with significant language-specific behavior, a threshold is defined based on the 95<sup>th</sup> percentile of the unnormalized activation probabilities:

$$\text{Threshold} = \text{Quantile}_{0.95}(p_{i,j}^l | \forall i, j, l). \quad (3.14)$$

A neuron  $(i, j)$  is considered for language-specific analysis only if:

$$\exists l \in \{1, \dots, k\} \text{ such that } p_{i,j}^l > \text{Threshold}. \quad (3.15)$$

**Step 5: Select Top Neurons Based on LAPE.** Only 1% neurons with the lowest LAPE values are selected for language-specific analysis, where  $m$  is defined as:

$$m = \frac{1 \times 4d \times L}{100}. \quad (3.16)$$

The top  $m$  neurons are identified by sorting the LAPE values and selecting the indices of the  $m$  smallest values:

$$\text{Top}_m(\text{Lang\_Set}) = \text{argsort\_ascending}_m \{ \text{LAPE}(i, j) | \forall i, j, \exists l \text{ s.t. } p_{i,j}^l > \text{Threshold} \}. \quad (3.17)$$

This ensures that only the most language-specific neurons are retained for further interpretation.

**Step 6: Identify Language-Specific Neuron Sets.** For each neuron  $(i, j)$  in the selected set, the language-specific set is defined as:

$$\text{Lang\_Spec\_Set}(i, j) = \{l \mid p_{i,j}^l > \text{Threshold}, (i, j) \in \text{Lang\_Set}\}. \quad (3.18)$$

A neuron is considered specific to a subset of languages if:

$$\text{Lang\_Spec\_Set}(i, j) \subseteq \{1, 2, \dots, k\}. \quad (3.19)$$

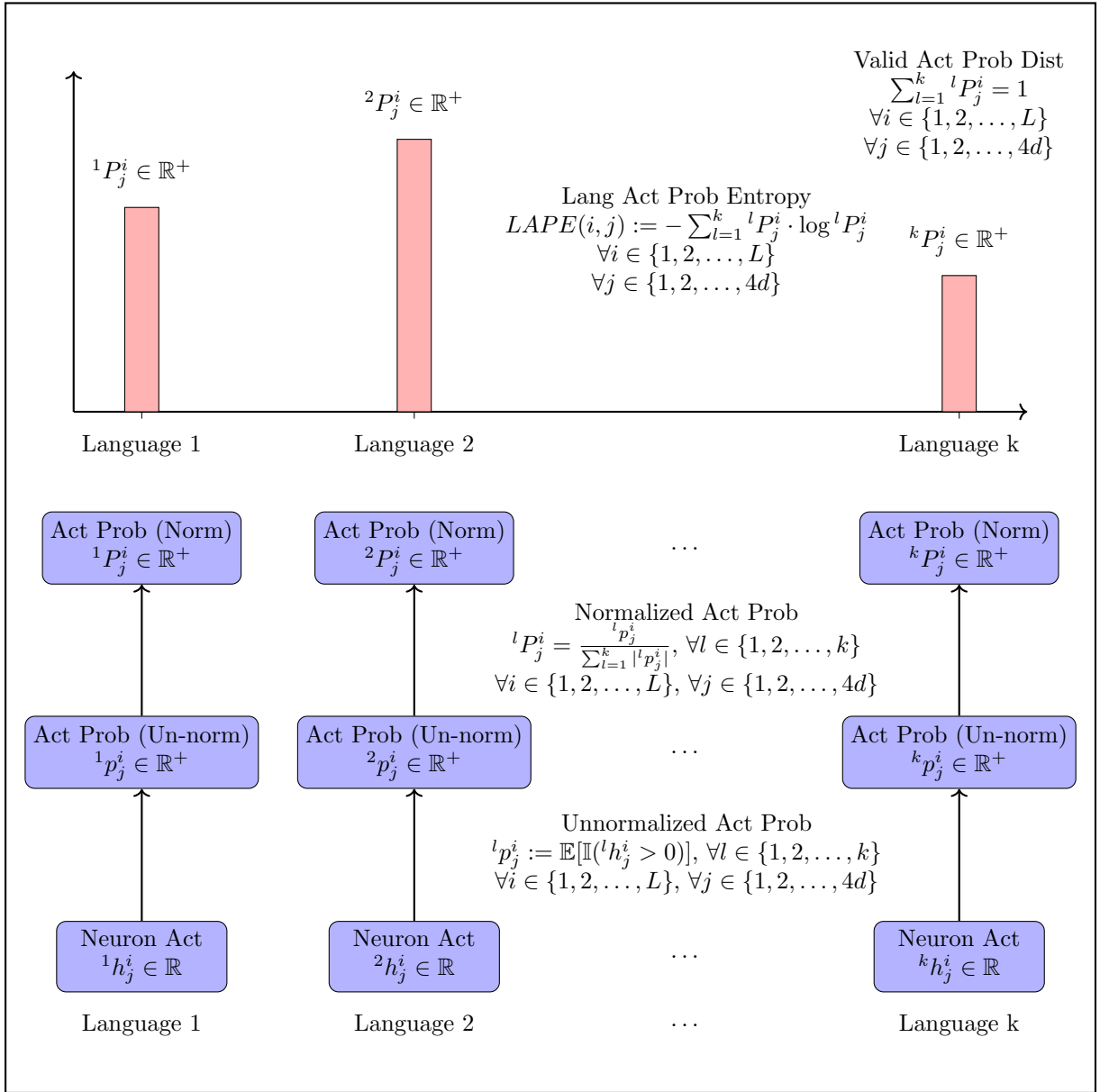


FIGURE 3.2: Procedure of computing LAPE for a neuron.

### 3.4.2 Activation Based Methods

**Absolute Activation:** The absolute activation metric measures the mean absolute activation of each neuron  $(i, j)$  in the  $i^{\text{th}}$  layer across all tokens in a sequence for a language  $l$ . This is computed as:

$$r_{i,j}^l = \mathbb{E} [|h_{i,j}^l|], \quad (3.20)$$

where  $h_{i,j}^l$  is the activation of neuron  $(i, j)$  for language  $l$ , and the expectation is taken over all tokens and sequences. This metric identifies neurons with consistently high absolute activations, which are indicative of their overall importance in representing language features.

**Activation Probability Zero:** The activation probability zero metric measures the fraction of times a neuron  $(i, j)$  has a positive activation for a language  $l$ . It is defined as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.21)$$

where  $\mathbb{I}$  is the indicator function that returns 1 if  $h_{i,j}^l > 0$ , and 0 otherwise. This metric captures the frequency of a neuron's activation, providing insights into its contribution to language-specific representations.

**Activation Probability Mean:** The activation probability mean measures the probability of a neuron's activation exceeding the mean activation value for its layer,  $\mu_{i,j}^l$ , for a language  $l$ . It is computed as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \mu_{i,j}^l)], \quad (3.22)$$

where  $\mu_{i,j}^l$  represents the average activation of neuron  $(i, j)$  in the  $i^{\text{th}}$  layer for language  $l$ . This metric identifies neurons with activations that are consistently above the average, highlighting neurons that are highly responsive to language-specific features.

**Activation Probability 90p:** The activation probability 90p measures the probability of a neuron's activation exceeding the 90th percentile activation value for its layer,  $\text{P90}_{i,j}^l$ , for a language  $l$ . This is given by:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > \text{P90}_{i,j}^l)], \quad (3.23)$$

where  $\text{P90}_{i,j}^l$  represents the 90th percentile of the activations of neuron  $(i, j)$  in the  $i^{\text{th}}$  layer for language  $l$ . This metric identifies neurons that exhibit rare but significant activations, which may correspond to specialized processing for certain linguistic features.

Language-specific neurons are identified based on activation values computed across different languages. For each language, activations are collected, and neurons that exhibit the highest relevance are selected using the following process.

**Step 1: Compute Activation Relevance.** For each language  $l \in \{1, 2, \dots, k\}$ , compute the activation relevance values for all neurons  $(i, j)$  in the pretrained model. Let the relevance for neuron  $(i, j)$  in the  $i^{\text{th}}$  layer for language  $l$  be denoted as  $r_{i,j}^l$ . This relevance is calculated using one of the activation probability methods, such as:

$$r_{i,j}^l = \mathbb{E} [\mathbb{I}(h_{i,j}^l > 0)], \quad (3.24)$$

or other methods such as absolute activation or percentile-based thresholds, depending on the desired analysis.

**Step 2: Aggregate Relevance Across Layers and Neurons.** For each language  $l$ , the computed relevance values  $r_{i,j}^l$  are aggregated across all layers and neurons. The total number of layers is denoted by  $L$ , and the dimensionality of the neurons in each layer is  $4d$ . The relevance tensor for language  $l$  is represented as:

$$R^l = \{r_{i,j}^l \mid i \in \{1, \dots, L\}, j \in \{1, \dots, 4d\}\}. \quad (3.25)$$

**Step 3: Select Top Neurons.** For each language  $l$ , the top  $m$  neurons with the highest relevance values are selected. The value of  $m$  is determined as a small fraction of the total number of neurons in the model:

$$m = \frac{0.125 \times L \times 4d}{100}. \quad (3.26)$$

The top  $m$  neurons are identified by sorting the relevance values in  $R^l$  and selecting the indices of the  $m$  largest values:

$$\text{Top}_m(R^l) = \text{argsort\_descending}_m \{r_{i,j}^l \mid \forall i, j\}. \quad (3.27)$$

**Step 4: Map Neurons to Layers and Positions.** The indices of the top  $m$  neurons are mapped back to their corresponding layers and positions. For each language  $l$ , the selected neurons are represented as:

$$N^l = \{(i, j) \mid (i, j) \in \text{Top}_m(R^l)\}, \quad (3.28)$$

where  $N^l$  is the set of top  $m$  neurons for language  $l$ .

This process ensures that neurons with the highest language-specific relevance are identified for each language. These neurons can then be analyzed further to understand their behavior and contribution to language modeling tasks.

### 3.4.3 Task Neurons for a Language

In this method, we introduce an alternative method based on task-specific neurons computed from the task specific dataset (for example XNLI dataset). Unlike language-specific neurons, which are computed using activation patterns derived from Wikipedia sentences, the task neurons are computed by processing the training sentences from the task dataset for each respective language. This modification allows the identification of neurons that are not only language-specific but also task-specific, capturing finer granularity of representations relevant to the classification task.

**Activation of Task-Specific Neurons:** The activation pattern for task-specific neurons is derived as:

$$h_{ij}^{\text{task}} = \text{stat}_{ij}^{\text{task}}, \quad \forall \text{stat} \in \{\mu, p_{90}, p_{75}, p_{50}, p_{25}, p_{10}, p_5\}, \quad (3.29)$$

where:

- $h_{ij}^{\text{task}}$  represents the activation for the  $i$ -th layer and  $j$ -th neuron under task-specific conditions.
- $\text{stat}_{ij}^{\text{task}}$  denotes the statistical activation computed over all task specific training sentences in a given language.
- The set  $\{\mu, p_{90}, p_{75}, p_{50}, p_{25}, p_{10}, p_5\}$  represents the mean, 90th percentile, 75th percentile, and other quantile thresholds used to analyze neuron activation levels.

Once the activation is computed for a task, then we can apply the LAPE or any activation based methods to compute the task specific neurons.

### 3.5 Neuron Fine-Tuning using LoRA for a Task

**LoRA (Low-Rank Adaptation):** LoRA [Hu et al. (2021)] introduces a mechanism to fine-tune a subset of parameters in neural networks, focusing on reducing computational overhead. For our application, we utilize LoRA to specifically fine-tune targeted neurons in the MLP layer while freezing others. This method ensures that the pretrained parameters  $W_{pt}$  remain untouched, and  $\Delta W$  represents the modifications applied to the network during fine-tuning.

**Fine-Tuning with LoRA AB Decomposition.** The matrix  $\Delta W$  is expressed as:

$$\Delta W = A \cdot B, \quad (3.30)$$

where  $A \in \mathbb{R}^{d_{in} \times r}$  and  $B \in \mathbb{R}^{r \times d_{out}}$  are low-rank matrices. The rank  $r$  is typically much smaller than  $d_{in}$  or  $d_{out}$ , reducing the number of trainable parameters. Masks are applied to  $A$  and  $B$  to control the fine-tuning of specific neurons:

$$A = \text{masked\_A} \cdot A_t, \quad B = \text{masked\_B} \cdot B_t, \quad (3.31)$$

where  $\text{masked\_A}$  and  $\text{masked\_B}$  are binary masks, and  $A_t$  and  $B_t$  are trainable parameters.

#### Description of Masked Matrices:

The binary masks  $\text{masked\_A}$  and  $\text{masked\_B}$  are defined to control the fine-tuning process by selectively freezing specific neurons.

$$\text{masked\_A} \in \mathbb{R}^{d_{in} \times r}, \quad \text{masked\_B} \in \mathbb{R}^{r \times d_{out}},$$

where  $d_{in}$  and  $d_{out}$  are the input and output dimensions of the layer, and  $r$  is the rank of the LoRA decomposition.

**Masked Matrix  $\text{masked\_A}$ :**  $\text{masked\_A}$  is defined as:

$$[\text{masked\_A}]_{ij} = 1, \quad \forall i \in \{1, \dots, d_{in}\}, \forall j \in \{1, \dots, r\}. \quad (3.32)$$

This indicates that no masking is applied to the rows of  $A$ , and all elements of  $\text{masked\_A}$  are set to 1.

**Masked Matrix  $\text{masked\_B}$ :** For  $\text{masked\_B}$ , specific columns corresponding to frozen neurons are set to 0. For a neuron index  $j_f \in \text{Frozen Neurons}$ , the mask is defined as:

$$[\text{masked\_B}]_{ij} = \begin{cases} 0, & \text{if } j = j_f, j_f \in \text{Frozen Neurons}, \\ 1, & \text{otherwise.} \end{cases} \quad (3.33)$$

**Final Masking Condition:** The effective masked matrices for  $A$  and  $B$  are:

$$A = \text{masked\_A} \cdot A_t, \quad B = \text{masked\_B} \cdot B_t, \quad (3.34)$$

where  $A_t \in \mathbb{R}^{d_{in} \times r}$  and  $B_t \in \mathbb{R}^{r \times d_{out}}$  are the trainable LoRA matrices.

This masking ensures that updates to the frozen neurons are completely suppressed, while other neurons remain trainable during fine-tuning.

**Frozen Neurons.** To freeze specific neurons, columns of  $B$  corresponding to the frozen neurons are masked with zeros. For neuron indices  $j$  in the set of frozen neurons, we ensure:

$$\Delta W_{i,j} = 0, \quad \forall i. \quad (3.35)$$

This prevents any updates to these neurons during fine-tuning.

**Final Output.** The fine-tuned output of the MLP layer is given by:

$$y = x \cdot W + x \cdot \Delta W, \quad (3.36)$$

where  $W$  is the original weight matrix, and  $\Delta W$  introduces targeted updates through LoRA.

**Targeted Language Neurons.** Language-specific neurons are identified and fine-tuned using LoRA. This targeted approach ensures that only the relevant neurons for a particular language are adapted while the others remain unchanged. For the frozen neurons,  $\Delta W$  remains zero:

$$\Delta W_{i,j} = 0, \quad \text{if } j \in \text{Frozen Neurons}. \quad (3.37)$$

**Additional Fine-Tuning.** Besides fine-tuning the MLP layer, the attention mechanisms ( $q, k, v, o$  projections) and the task-specific classifier head are always fine-tuned using LoRA. This ensures the overall task-specific performance is preserved while leveraging the language neurons for targeted adaptation. Figure 3.3 shows the procedure of finetuning target neurons using LoRA.

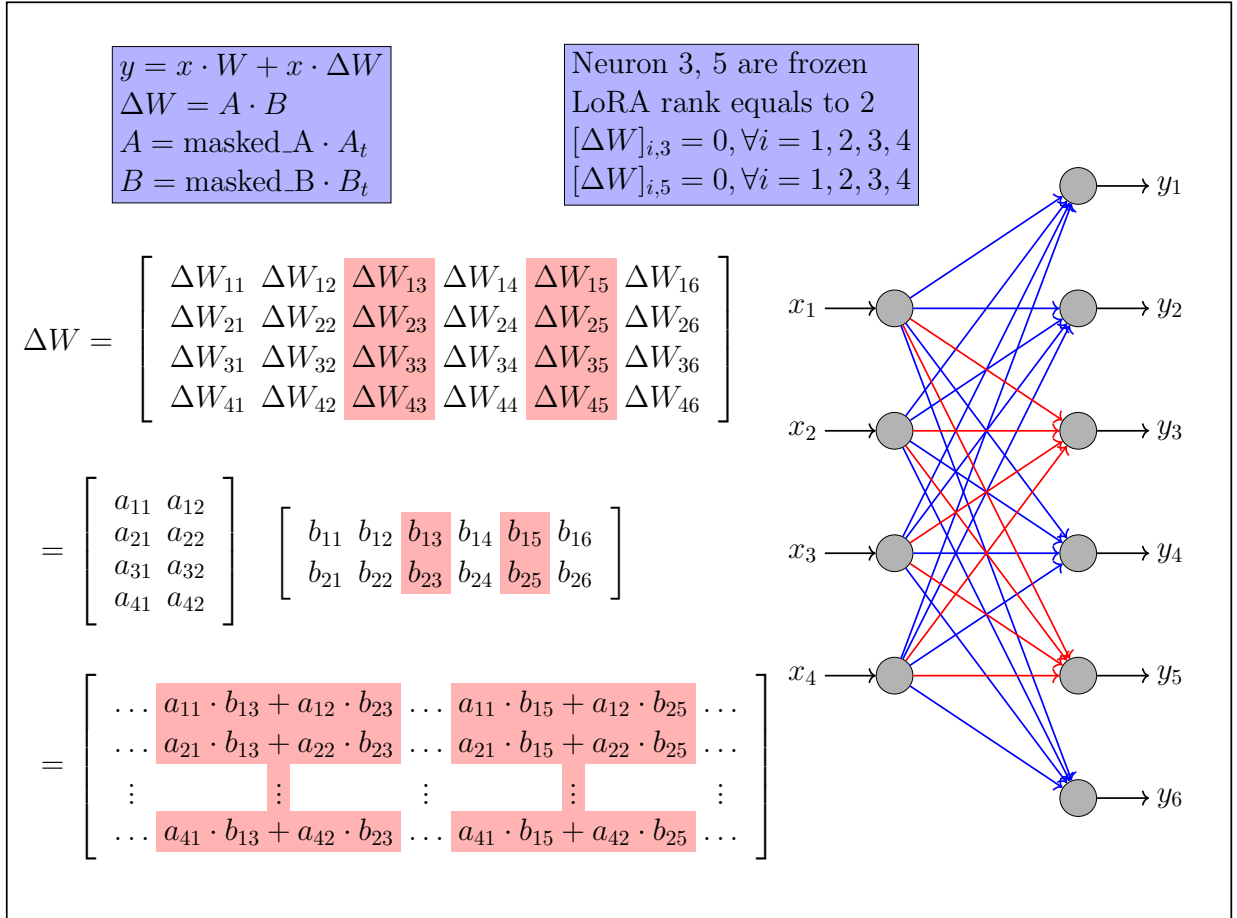


FIGURE 3.3: LoRA fine tuning method for target neurons.

# Chapter 4

## Experiments

### 4.1 Task and Dataset: XNLI

The Cross-lingual Natural Language Inference (XNLI) [Conneau and Others (2018)] task is a benchmark designed to evaluate the capability of large language models (LLMs) to perform reasoning across multiple languages. The goal is to classify the relationship between a premise and a hypothesis into one of three categories: *entailment*, *neutral*, or *contradiction*. This task tests the ability of the model to generalize semantic relationships beyond the English language, making it a challenging benchmark for multilingual models.

**Dataset Overview:** The XNLI dataset contains annotated sentence pairs across 15 languages. In our experiments, we focus on English (*en*) as the source language and Vietnamese (*vi*), Hindi (*hi*) and Urdu (*ur*) as target languages. The dataset for each language can be represented as:

$$\mathcal{D}_l = \{(x_p^i, x_h^i, y^i) \mid x_p^i, x_h^i \in \mathcal{X}_l, y^i \in \{0, 1, 2\}\}, \quad (4.1)$$

where  $x_p^i$  and  $x_h^i$  denote the premise and hypothesis sentences in language  $l$ , and  $y^i$  is the corresponding label indicating the relationship.

**Tokenization and Preprocessing:** Each sentence pair  $(x_p^i, x_h^i)$  is concatenated with a special token to form the input:

$$x^i = x_p^i + [\text{EOS}] + x_h^i, \quad (4.2)$$

where  $[\text{EOS}]$  represents the end-of-sequence token. The concatenated input is tokenized into a sequence of tokens  $\{t_1, t_2, \dots, t_T\}$  with a maximum sequence length  $T = 512$ .

**Train and Test Splits:** The dataset is divided into training and testing subsets for each language:

$$\mathcal{D}_l = \mathcal{D}_l^{\text{train}} \cup \mathcal{D}_l^{\text{test}}, \quad \mathcal{D}_l^{\text{train}} \cap \mathcal{D}_l^{\text{test}} = \emptyset. \quad (4.3)$$

For the experiments, a fraction  $\mathcal{F} \subset \mathcal{D}_l^{\text{train}}$  is used to train the model. The size of  $\mathcal{F}$  is determined as:

$$|\mathcal{F}| = \text{frac} \times |\mathcal{D}_l^{\text{train}}|, \quad (4.4)$$

where  $\text{frac}$  is the fraction of the training dataset used.

**Multilingual Challenges:** In the XNLI task, transferring knowledge from English to other languages like Vietnamese and Hindi requires the model to overcome both semantic and structural differences in language. This transfer is tested by evaluating the zero-shot performance on the target languages without directly training on their data. The objective of this task is to fine-tune LLaMA3, a large language model, on the XNLI dataset to improve cross-lingual inference capabilities. The experiments are designed to test the effectiveness of various fine-tuning setups and analyze the role of language-specific neurons in improving performance across multiple languages.

## 4.2 Baseline Experiments

The baseline experiments establish a foundation for evaluating the performance of different fine-tuning setups on the XNLI task. We include two baselines: zero-shot evaluation of a model fine-tuned on English (*en*) XNLI data and direct fine-tuning on the target language (*vi*) XNLI data. These baselines provide a comparative framework to assess the impact of fine-tuning specific language neurons using LoRA.

**Zero-Shot Baseline:** The first baseline involves fine-tuning the pretrained model on the English XNLI dataset. Let  $\mathcal{M}_{\text{en}}$  denote the model fine-tuned on English data. The zero-shot evaluation involves testing  $\mathcal{M}_{\text{en}}$  on Vietnamese XNLI datasets without further training. The evaluation metric, accuracy, is defined as:

$$\text{Accuracy} = \frac{\sum_{i=1}^N \mathbb{I}(\hat{y}^i = y^i)}{N}, \quad (4.5)$$

where  $N$  is the total number of test examples,  $y^i$  is the ground truth label,  $\hat{y}^i$  is the predicted label, and  $\mathbb{I}(\cdot)$  is the indicator function.

**Direct Fine-Tuning Baseline:** The second baseline involves directly fine-tuning the pretrained model on the Vietnamese XNLI dataset. Let  $\mathcal{M}_{\text{vi}}$  denote the model fine-tuned on Vietnamese data. This baseline establishes an upper bound for performance on the Vietnamese XNLI task, as the model is explicitly trained on the target language data. We again use the accuracy as evaluation metric.

## 4.3 Experiment Goals

The primary objective of this work is to improve the zero-shot performance of the model on the target language, such as Vietnamese (*vi*), without utilizing any target language training data. This constraint is critical to ensure that the model remains agnostic to the specific characteristics of the target language during training, thereby simulating a real-world cross-lingual transfer learning scenario.

**Baseline Comparison:** In this setup, we consider two baselines:

- **Baseline 1: Zero-Shot Performance on *vi* using English Training Data.** This baseline involves fine-tuning the pretrained model on the English (*en*) XNLI dataset ( $\mathcal{M}_{\text{en}}$ ) and evaluating its zero-shot performance on the Vietnamese (*vi*) test set. This represents the starting point of cross-lingual transfer without any modifications to the model architecture or targeted interventions.
- **Baseline 2: Performance on *vi* using Direct Fine-Tuning on *vi* Data.** This baseline involves directly fine-tuning the pretrained model on the Vietnamese (*vi*) XNLI dataset ( $\mathcal{M}_{\text{vi}}$ ). It provides an upper bound on performance since it leverages target language training data.

**Objective Constraints:** While the ultimate goal is to improve the zero-shot performance on the *vi* test set, it is imperative to operate under the following constraints:

1. No target language (*vi*) training data can be used during the fine-tuning process.
2. Only the English (*en*) XNLI training data is available for model training and intervention.
3. The interventions must be limited to model surgery techniques, such as fine-tuning specific neurons or modifying activation patterns.

**Target Metric:** The improvement in zero-shot performance is quantified as:

$$\Delta_{\text{zs}} = \text{Accuracy}(\mathcal{M}_{\text{experiment}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.6)$$

where  $\mathcal{M}_{\text{experiment}}$  refers to the model modified through various intervention strategies, and  $\mathcal{M}_{\text{en}}$  refers to the baseline model fine-tuned on *en*.

**Goal:** The overarching goal is to maximize  $\Delta_{\text{zs}}$ , thereby surpassing the performance of Baseline 1 ( $\mathcal{M}_{\text{en}}$ ) through innovative techniques such as:

- Fine-tuning language-specific neurons identified for the target language (*vi*) using only the *en* training data.
- Utilizing statistical activation patterns (*e.g.*, *mean*, *quantile-based thresholds*) to intervene in neuron activations during inference.
- Applying LoRA (Low-Rank Adaptation) to selectively fine-tune specific components such as *qkv* projections and classification heads while freezing other parts of the model.

**Limitations:** By design, the performance of any intervention cannot exceed Baseline 2 ( $\mathcal{M}_{\text{vi}}$ ), as that baseline uses direct training data from the target language. However, our approach aims to close the gap between Baseline 1 ( $\mathcal{M}_{\text{en}}$ ) and Baseline 2 ( $\mathcal{M}_{\text{vi}}$ ) as much as possible through zero-shot strategies.

## 4.4 Targeted Neuron Fine-tuning Experiment

This section outlines the experimental setups designed to improve the zero-shot performance of the model on the Vietnamese (*vi*) XNLI dataset. Four configurations are explored, each using different strategies to fine-tune the model, while adhering to the constraint of using only English (*en*) training data during fine-tuning. The experiments aim to utilize targeted interventions to improve performance by selectively tuning neurons and applying LoRA-based adaptations.

**Setup A: Standard Fine-Tuning with LoRA.** In this setup, the pretrained model is fine-tuned on the English (*en*) XNLI dataset using LoRA. The fine-tuning is applied to the query ( $q$ ), key ( $k$ ), value ( $v$ ), and output ( $o$ ) projection matrices ( $W_q^i, W_k^i, W_v^i, W_o^i$ ) and the classification head. The resulting model is evaluated on the Vietnamese (*vi*) dataset in a zero-shot setting. Additionally, the *vi*-specific activations are substituted by statistical activation patterns such as mean ( $\mu$ ), 90<sup>th</sup> percentile, and others, defined as:

$$\hat{h}_j^{i,vi} = \text{stat}_j^{i,vi}, \quad \forall \text{stat} \in \{\text{mean}, 90p, 95p, 75p, 5p, 10p, 25p, 0\}, \quad (4.7)$$

where  $\hat{h}_j^{i,vi}$  represents the substituted activation for *vi* language for neuron  $(i, j)$  in the MLP block.

**Setup B: Fine-Tuning with LoRA and English Neurons.** This setup builds on Setup A by incorporating fine-tuning of English-specific neurons identified in the MLP block. Let  $\mathcal{N}_{\text{en}}$  denote the set of English-specific neurons. During fine-tuning, the LoRA modifications ( $\Delta W$ ) are applied to both the query-key-value-output projections and the English neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en}}. \quad (4.8)$$

This targeted approach allows the model to specialize in English while preserving the generalization capacity of other neurons.

**Setup C: Fine-Tuning with LoRA and Target Language Neurons.** In this configuration, the focus shifts to fine-tuning target language such as Vietnamese-specific neurons ( $\mathcal{N}_{\text{vi}}$ ). During fine-tuning, the LoRA modifications ( $\Delta W$ ) are applied to both the query-key-value-output projections and the Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{vi}}. \quad (4.9)$$

This ensures that only the Vietnamese-specific neurons are fine-tuned while others remain untouched.

**Setup D: Fine-Tuning with LoRA and Both English & Target Language Neurons.** This setup combines the interventions from Setup B and Setup C, targeting both English and Vietnamese neurons. Let  $\mathcal{N}_{\text{en,vi}} = \mathcal{N}_{\text{en}} \cup \mathcal{N}_{\text{vi}}$  represent the union of English and Vietnamese neurons. During fine-tuning, the LoRA modifications ( $\Delta W$ ) are applied to both the query-key-value-output projections and both the English & Vietnamese neurons, ensuring that:

$$\Delta W_j^i = 0, \quad \forall j \notin \mathcal{N}_{\text{en,vi}}. \quad (4.10)$$

This strategy allows the model to simultaneously adapt to English and Vietnamese language-specific representations.

**Comparative Analysis:** After fine-tuning, the zero-shot performance on Vietnamese (*vi*) XNLI data is measured using the accuracy metric. The performance improvements achieved through the setups are quantified by:

$$\Delta_{\text{ZS}} = \text{Accuracy}(\mathcal{M}_{\text{setup}}) - \text{Accuracy}(\mathcal{M}_{\text{en}}), \quad (4.11)$$

where  $\mathcal{M}_{\text{setup}}$  refers to the model fine-tuned using one of the experimental setups. These improvements are compared against the baselines to evaluate the efficacy of the targeted neuron fine-tuning strategies.

## 4.5 Targeted Neuron Fine-Tuning Configurations

**Training Overview:** The training of the XNLI task was conducted using the LoRA-based fine-tuning strategy, leveraging the Llama3.1 model. The experimental setup includes both English and multilingual datasets, with targeted interventions in language-specific neurons. The primary objective was to fine-tune the model while freezing specific subsets of neurons based on language-specific activations.

**Model and Quantization:** The Llama3 model was employed with 4-bit quantization, utilizing the nf4 quantization type for memory efficiency. The compute precision for quantization was set to bfloat16 with double quantization enabled. This configuration ensures minimal computational overhead without compromising the model's performance.

**Dataset:** The dataset used for training and evaluation was the **XNLI Dataset**. The dataset was tokenized using the pre-trained tokenizer corresponding to the Llama3 model, with a maximum context length of 512 tokens. The training dataset utilized 25% of the available data, while 10% was reserved for evaluation.

### Training Hyperparameters:

- **Epochs:** Training was conducted for a single epoch, divided into steps calculated based on the batch size and dataset size.
- **Batch Size:** A batch size of 8 was used for both training and evaluation, with gradient accumulation steps set to 1.
- **Learning Rate:** The initial learning rate was set to  $1 \times 10^{-5}$ , with a linear warm-up scheduler for the first 1% of steps.
- **Optimization:** The optimizer used was AdamW with  $\beta_1 = 0.95$ ,  $\beta_2 = 0.999$ , weight decay of 0.1, and a gradient clipping norm of 10.0.
- **LoRA Configuration:** The LoRA rank was set to 8, and the scaling factor ( $\alpha$ ) was set to 16.

# Chapter 5

## Analysis and Results

### 5.1 LAPE Neuron Analysis

#### 5.1.1 Language Sets

In this section, we define the different language sets used for evaluating the Language Activation Probability Entropy (LAPE) in the context of our experiments. The language sets are constructed to capture varying linguistic and typological characteristics across a diverse group of languages. The experimental results on these sets aim to analyze the impact of multilinguality on language-specific neuron activations.

**Set 1: Core Languages.** This set includes the core languages commonly supported by pretrained language models:

$$\text{Set 1} = \{\text{en, fr, es, vi, id, ja, zh}\}.$$

These languages span across major linguistic families and represent both high-resource and typologically distinct languages.

**Set 2: Extended Core with Indian Languages.** To incorporate a mix of Western and Indian languages, this set is defined as:

$$\text{Set 2} = \{\text{en, fr, vi, zh, bn, hi, te, mr}\}.$$

Here, bn (Bengali), hi (Hindi), te (Telugu), and mr (Marathi) represent Indian languages, providing an interesting contrast to the Western languages in the set.

**Set 3: Indian Subset.** This set focuses exclusively on Indian languages:

$$\text{Set 3} = \{\text{bn, hi, ta, te, mr, ur, kn, ml, pa}\},$$

where ta (Tamil), ur (Urdu), kn (Kannada), ml (Malayalam), and pa (Punjabi) are included to cover a broader typological spectrum of Indian languages.

**Set 4: Core and Extended Languages.** Combining the core and extended sets, this set is defined as:

$$\text{Set 4} = \{\text{en, fr, es, vi, id, ja, zh, bn, hi, te}\}.$$

This allows for the comparison of results across a comprehensive multilingual setup.

**Set 5: All Languages.** This set includes all the languages in the experimental framework:

$$\text{Set 5} = \{\text{en, fr, es, vi, id, ja, zh, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

**Set 6: Indian Language Dominant.** This set focuses on Indian languages while retaining a few globally dominant ones:

$$\text{Set 6} = \{\text{en, bn, hi, ta, te, mr, ur, kn, ml, pa}\}.$$

The diversity of the language sets ensures a thorough evaluation of language-specific neuron behavior. By analyzing LAPE across these sets, we aim to uncover insights into the multilingual capabilities of the pretrained model and the role of specific languages in driving neuron activations.

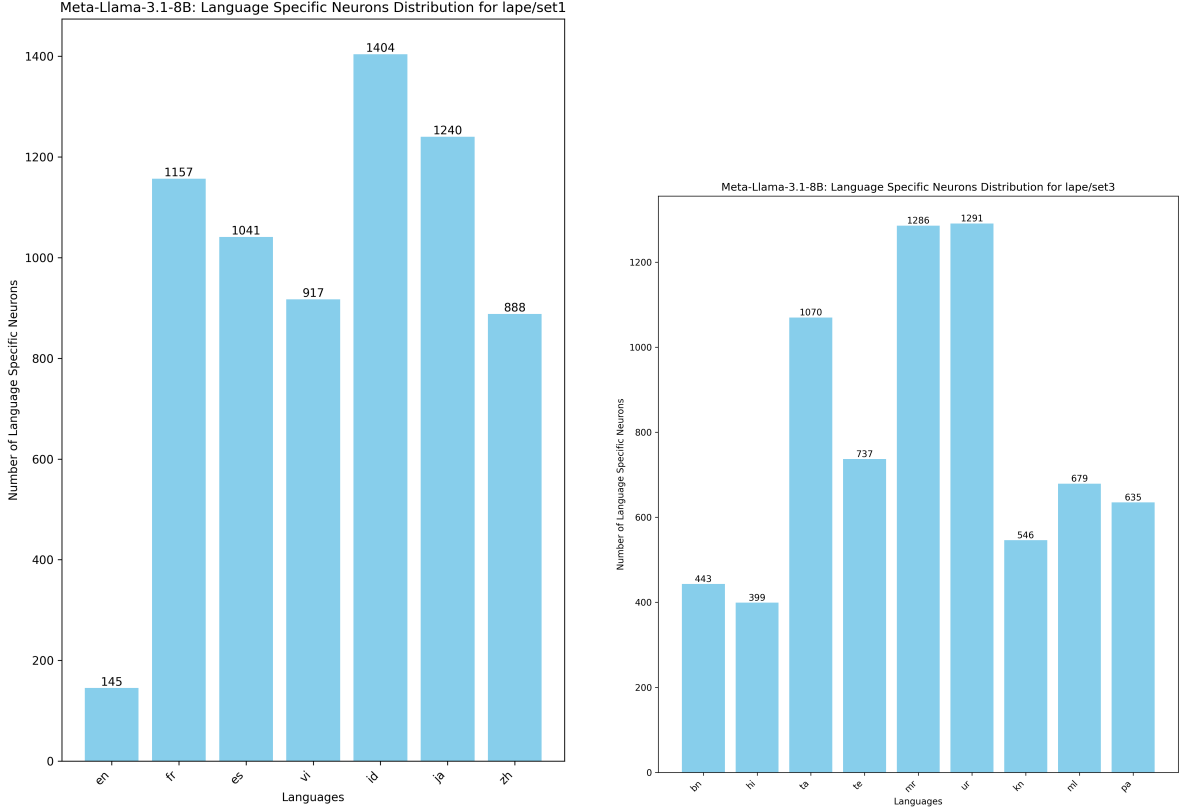
### 5.1.2 Language Neuron Distribution

The Language Activation Probability Entropy (LAPE) method enables the identification of language-specific neurons in large language models. This section analyzes the distribution of language-specific neurons across different languages, their overlap, and their distribution across model layers for the selected sets of languages, namely **Set1** and **Set3**. These analyses are crucial for understanding how linguistic diversity is represented within the model and for designing strategies for targeted interventions.

**Language-Specific Neuron Counts:** The number of language-specific neurons for each language in **Set1** and **Set3** is illustrated in Figure 5.1. The LAPE method assigns neurons to languages based on their activation probabilities. Formally, the number of neurons for a language  $l$  is computed as:

$$N_l = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang\_Set}), \quad (5.1)$$

where  $N_l$  is the total number of neurons assigned to language  $l$ ,  $p_{i,j}^l$  denotes the un-normalized activation probability for a language neuron  $(i, j) \in \text{Lang\_Set}$  and language  $l$ , and  $\mathbb{I}$  is the indicator function. The threshold ensures that only the most prominent neurons are included.



**Set1: Language Neuron Distribution    Set3: Language Neuron Distribution**

FIGURE 5.1: Language-specific neuron distributions for LAPE method for Set1 and Set3.

From Figure 5.1, it is evident that certain languages, such as *id*, *ja*, *ur*, and *mr*, have higher neuron counts, suggesting that the model dedicates more capacity to languages with rich morphology or syntactic structures. Conversely, *en* consistently has the fewest neurons, likely due to its large amount of training data.

**Neuron Overlap Between Languages:** Figure 5.2 illustrates the overlap of language-specific neurons between languages in Set1 and Set3. The overlap is formally calculated as:

$$O_{l_1, l_2} = \sum_{i=1}^L \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^{l_1} > \text{Threshold}) \cdot \mathbb{I}(p_{i,j}^{l_2} > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang\_Set}), \quad (5.2)$$

where  $O_{l_1, l_2}$  represents the number of neurons shared between languages  $l_1$  and  $l_2$ .

Figure 5.2 highlights that languages with shared syntactic or morphological properties, such as *id* and *vi* in Set1, exhibit higher overlap. In contrast, languages like *en* and *fr* have minimal overlap with other languages, underscoring their unique linguistic structures.

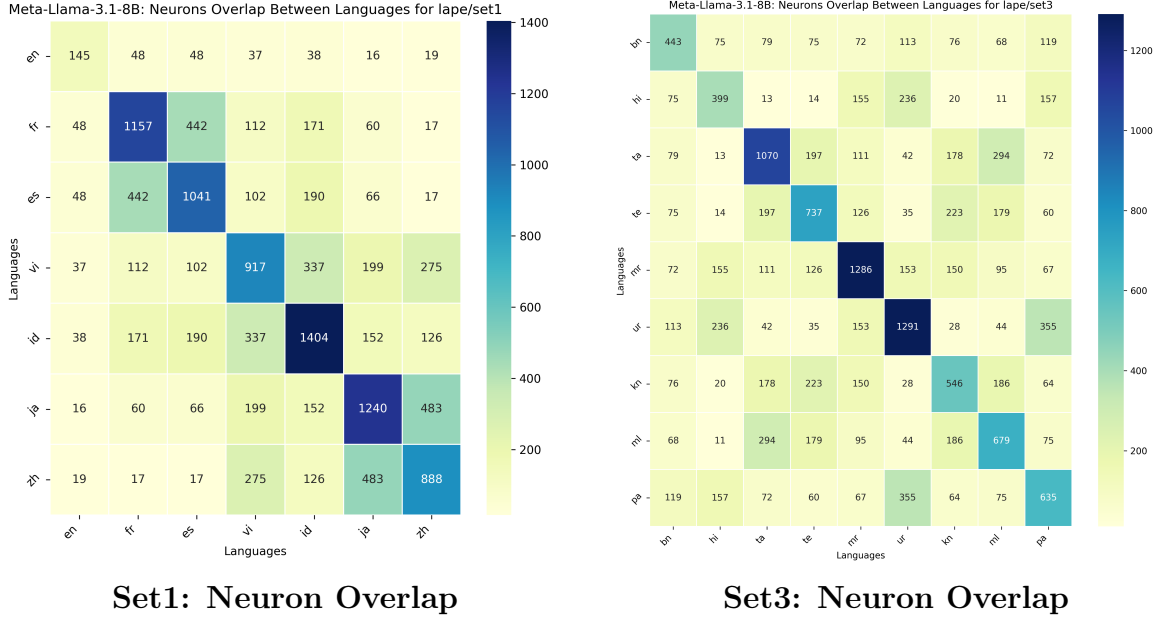


FIGURE 5.2: Neuron overlap between languages for LAPE method for Set1 and Set3.

**Layer-Wise Neuron Distribution:** The layer-wise distribution of language-specific neurons for Set1 and Set3 is shown in Figure 5.3. The distribution is computed as:

$$L_l(i) = \sum_{j=1}^{4d} \mathbb{I}(p_{i,j}^l > \text{Threshold}) \cdot \delta((i, j) \in \text{Lang\_Set}), \quad (5.3)$$

where  $L_l(i)$  denotes the number of neurons in layer  $i$  associated with language  $l$ .

Figure 5.3 reveals that language-specific neurons are concentrated in the upper layers (e.g., Layers 27-30), aligning with the notion that deeper layers capture more abstract and language-specific features. Lower layers exhibit fewer neurons, as they primarily encode shared linguistic features.

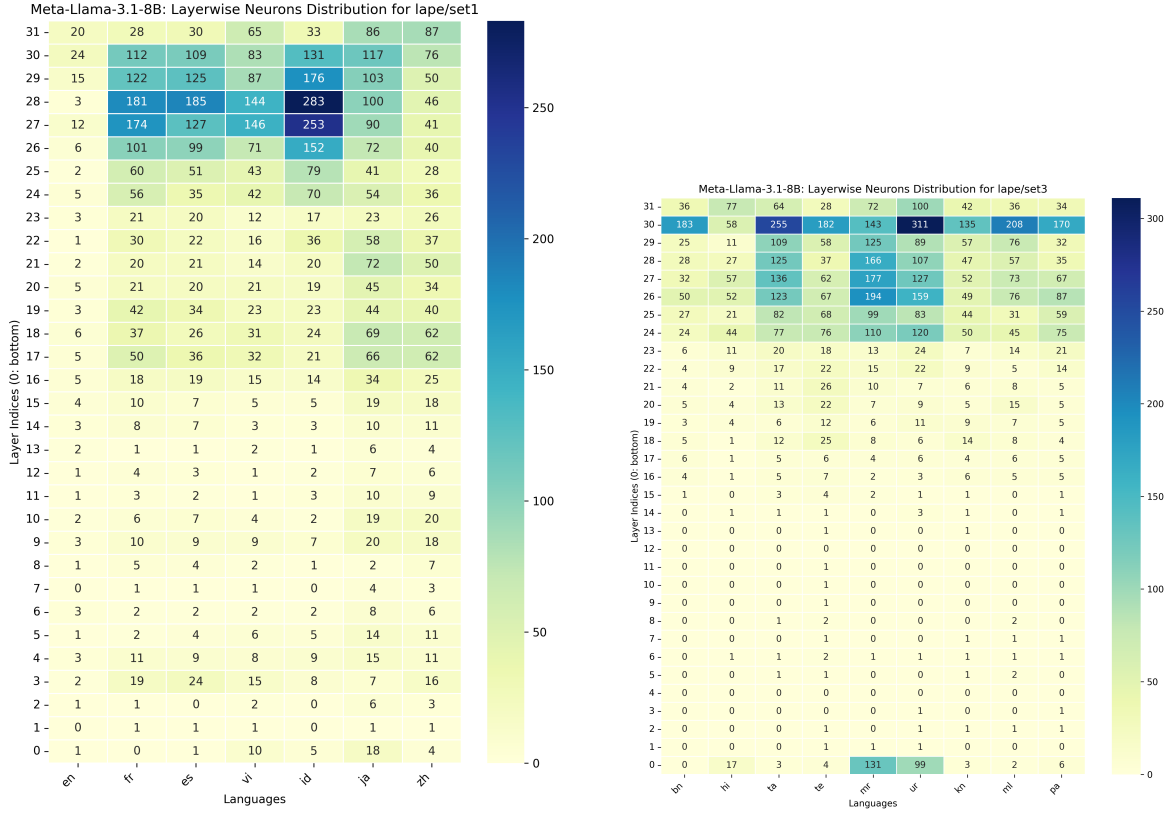
### 5.1.3 Perplexity Change

**Perplexity (PPX):** Perplexity is a widely used metric to evaluate the performance of language models. It measures how well a model predicts a sequence of tokens and is formally defined as the exponential of the average negative log-probability of the tokens in a sequence.

Mathematically, for a sequence of tokens  $\mathbf{x} = (x_1, x_2, \dots, x_T)$ , the perplexity is given by:

$$\text{PPX}(\mathbf{x}) = \exp \left( -\frac{1}{T} \sum_{t=1}^T \log P(x_t | x_{<t}) \right), \quad (5.4)$$

where:



Set1: Layer-wise Neuron Distribution    Set3: Layer-wise Neuron Distribution

FIGURE 5.3: Layer-wise neuron distributions for LAPE method for Set1 and Set3.

- $T$  is the length of the sequence.
- $P(x_t | x_{<t})$  is the probability assigned by the model to the token  $x_t$ , given the preceding tokens  $x_{<t} = (x_1, x_2, \dots, x_{t-1})$ .

Perplexity can be understood as the average branching factor of the model in predicting tokens. A lower perplexity indicates that the model is more confident in its predictions, while a higher perplexity suggests greater uncertainty.

For a dataset  $\mathcal{D}$  consisting of multiple sequences, the perplexity is computed as the mean of the per-sequence perplexities:

$$\text{PPX}(\mathcal{D}) = \exp \left( -\frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} \log P(x_t^{(i)} | x_{<t}^{(i)}) \right), \quad (5.5)$$

where  $N$  is the total number of sequences in the dataset, and  $T_i$  is the length of the  $i$ -th sequence.

**Perplexity Change (PPXC):** In the context of language modeling, perplexity serves as a key metric to measure the performance of a model. It quantifies how well a probability distribution

predicts a sample. The perplexity change  $PPXC(i, j)$  measures the change in perplexity for a target language  $j$  when an intervention is applied at a source language  $i$ . Formally, it is defined as:

$$PPXC(i, j) = PPX(j | \text{Intervention at } i) - PPX(j), \quad (5.6)$$

where  $PPX(j)$  denotes the original perplexity for language  $j$ , and  $PPX(j | \text{Intervention at } i)$  is the perplexity after performing an intervention at language  $i$ . Lower values of  $PPXC(i, j)$  indicate minimal interference, whereas higher values highlight a significant change in the model’s understanding of language  $j$  due to the intervention.

**Experimental Setup:** For this experiment, we used the Wikipedia dataset for each language. The experiments were conducted for two sets of languages: **Set1**, containing  $\{en, fr, es, vi, id, ja, zh\}$ , and **Set3**, containing  $\{bn, hi, ta, te, mr, ur, kn, ml, pa\}$ . The perplexity values were computed for each language, both with and without intervention at other languages.

**Analysis:** The Figure 5.4 illustrates the perplexity changes for Set1 and Set3. Each heatmap represents  $PPXC(i, j)$  values for different combinations of source and target languages. In all the experiments, the intervention means the values for the neuron activation is set as zero.

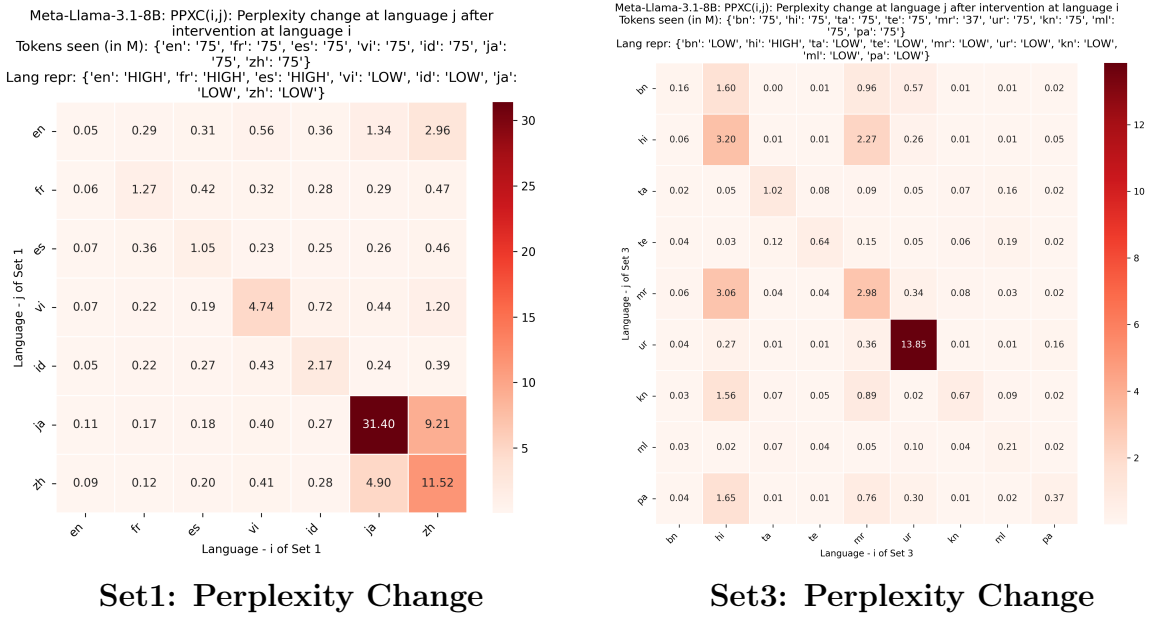


FIGURE 5.4: Perplexity change  $PPXC(i, j)$  for LAPE method for Set1 and Set3.

From Figure 5.4, we observe the following key trends:

- For **Set1**, interventions at low-resource languages such as *ja* lead to substantial perplexity changes for other low-resource languages (*zh*, *id*, *vi*), as indicated by high  $PPXC(i, j)$  values. This suggests that these languages share overlapping representations in the model.

- For **Set3**, interventions at *ur* result in significant perplexity changes for other low-resource languages like *ta*, *te*, highlighting the influence of high-representation languages on their low-representation counterparts.

## 5.2 Activation Probability 90p Neuron Analysis

### 5.2.1 Language Neuron Distribution

In this section, we analyze the distribution of language-specific neurons across the selected languages: *en*, *vi*, *hi*, and *ur*. Unlike LAPE, this approach is set-independent, meaning the results remain consistent regardless of the set of languages considered. Each language has the same number of neurons, precisely 573, allocated based on 0.0125% of the total number of neurons. This ensures fairness and uniformity in the analysis across different languages. The emphasis here is on examining the overlap and layer-wise distribution of these neurons. The overlap matrix, as shown in Figure 5.5, highlights the extent to which neurons are shared between languages. This indicates that while there is some degree of shared representation, the neurons largely retain their language specificity.

The layer-wise neuron distribution, depicted in Figure 5.5, provides insights into the depth at which language-specific features are encoded. For all languages, a significant concentration of neurons is observed in the lower layers (e.g., Layers 4–8). This suggests that fundamental linguistic features, such as syntax and basic semantics, are predominantly captured in the initial layers. In contrast, the middle and upper layers contain fewer language-specific neurons, indicating that these layers might focus more on higher-level abstractions or tasks that are less language-dependent.

This method demonstrates a significant advantage over LAPE as it ensures consistency and fairness across languages. Since the language-specific neurons are determined independently of the set of languages considered, the analysis does not suffer from the set-dependent biases inherent in LAPE.

### 5.2.2 Perplexity Change

The experimental results for the perplexity change (*PPXC*) are presented in this section. These results focus on analyzing the impact of interventions at one language (*i*) on the perplexity of another language (*j*) within the same experimental setup. The datasets used for these computations are derived from the Wikipedia corpus for the respective languages.

**Analysis:** From Figure 5.6, the following insights can be drawn:

- **Non-Diagonal Behavior:** The perplexity change values are not strictly diagonal, indicating that interventions in one language (*i*) can significantly influence the perplexity of another language (*j*). This behavior suggests a complex interplay between the language

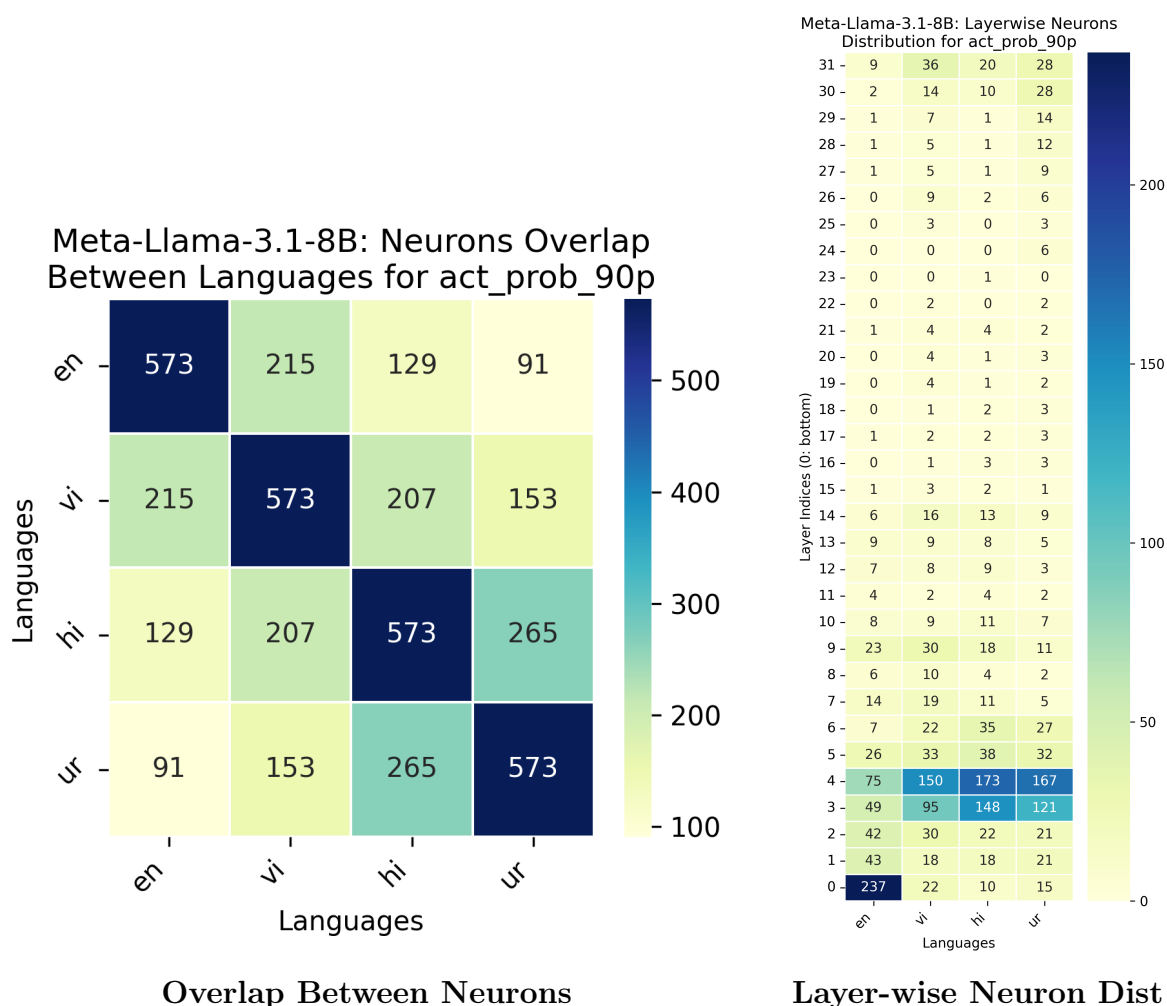
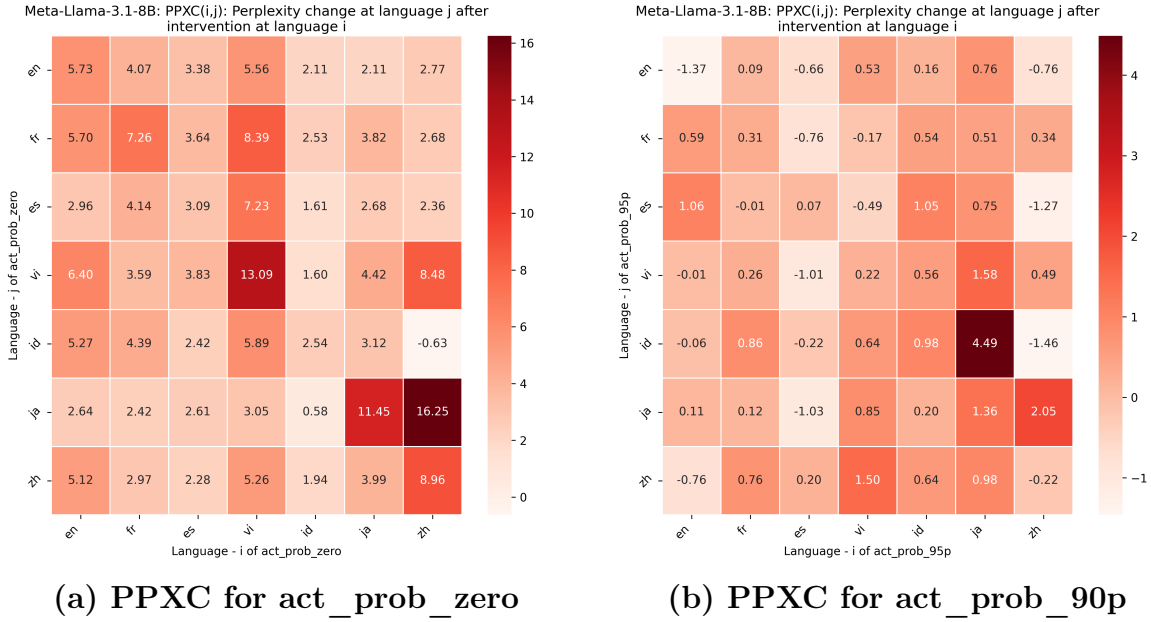


FIGURE 5.5: Overlap between neurons and layer-wise neurons distribution for activation probability 90p method.

representations within the model, where neurons may encode shared features across multiple languages.

- **Negative Values:** In some cases, the perplexity change values are negative, implying that deactivating certain neurons in one language can paradoxically improve the language capability of another. This counterintuitive result highlights the possibility that some neurons may act as bottlenecks, restricting the efficient representation of linguistic features in other languages.
- **Further Analysis Required:** The exact mechanisms underlying these phenomena are unclear and require further exploration. Factors such as the resource level of the languages, the layer-wise distribution of neurons, and the training dynamics of shared representations could play a critical role.

FIGURE 5.6: Perplexity change  $PPXC(i, j)$  for activation probability based methods.

## 5.3 Fine-Tuning Experiment Results on XNLI Task

### 5.3.1 Baseline Experiment Results

The baseline experiments establish a framework for evaluating different fine-tuning configurations on the XNLI task. Two key baselines are explored: *Zero-Shot Transfer* and *Direct Fine-Tuning*. Zero-shot transfer involves testing a model fine-tuned on English (*en*) on other languages without further adaptation. In contrast, direct fine-tuning entails training the model explicitly on the target language data. These baselines are compared to assess the effectiveness of targeted interventions in language-specific neurons.

| Baseline Type      | Train Lang | Eval Lang | No Int Accuracy |
|--------------------|------------|-----------|-----------------|
| Zero Shot          | en         | vi        | 0.8053          |
|                    | en         | hi        | 0.7501          |
|                    | en         | ur        | 0.7002          |
| Direct Fine-tuning | vi         | vi        | 0.8511          |
|                    | hi         | hi        | 0.8043          |
|                    | ur         | ur        | 0.6697          |

TABLE 5.1: Baseline results for the XNLI task for Llama 3.1.

**Insights from Baselines:** The results from Table 5.1 highlight several important observations:

1. **Zero-Shot Transfer Performance:** For languages such as Vietnamese (*vi*), zero-shot transfer using an English-fine-tuned model achieves an accuracy of 0.8053. This is a reasonable

performance considering the linguistic divergence between English and Vietnamese, demonstrating the model’s generalization capabilities.

**2. Direct Fine-Tuning Performance:** As expected, direct fine-tuning on the target language outperforms zero-shot transfer in all cases. For instance, fine-tuning directly on Vietnamese (*vi*) data yields an accuracy of 0.8511, surpassing the zero-shot accuracy by a significant margin.

**3. Performance Gap Between Languages:** The gap in accuracy between zero-shot transfer and direct fine-tuning is more pronounced for Vietnamese (*vi*) compared to Hindi (*hi*) and Urdu (*ur*). For *hi* and *ur*, the zero-shot performance (0.7501 and 0.7002, respectively) is relatively closer to the directly fine-tuned results (0.8043 and 0.6697, respectively). This suggests that the model may share common linguistic representations between English and these languages, leading to better transferability.

**4. Challenges with Low-Resource Languages:** The observation that direct fine-tuning for Urdu (*ur*) yields lower accuracy (0.6697) compared to zero-shot transfer (0.7002) suggests several plausible reasons. Firstly, the dataset used for fine-tuning might be limited in size or quality, leading to insufficient adaptation to the target language. The pretrained model’s underrepresentation of Urdu during training, combined with its unique linguistic and script characteristics, might also limit the efficacy of fine-tuning. In contrast, zero-shot transfer leverages cross-lingual embeddings and the model’s generalized multilingual capabilities, enabling better performance without the risk of overwriting or overfitting. This highlights the challenges of fine-tuning low-resource languages like Urdu and underscores the need for more robust and targeted approaches.

The baseline results establish that direct fine-tuning on target languages consistently improves performance. However, the competitive zero-shot performance for certain languages underscores the robustness of transfer learning. These findings provide a foundation for further evaluation of targeted interventions in language-specific neurons.

### 5.3.2 LAPE Experiment Results using Lang Specific Neurons

Table 5.2 summarizes the results of the fine-tuning experiments on the XNLI task. These experiments involve evaluating the zero-shot performance of the model on Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*) after targeted neuron fine-tuning and statistical interventions. The table reports the results (*No Int*) alongside the effects of interventions using statistical patterns (*Mean*, P75, P90, and P95).

#### Insights from Results of Intervention with Higher Quantiles:

- **No Intervention Performance:** The model (*No Int*) achieves robust zero-shot performance for all three languages. For instance, in *vi*, the baseline accuracy is highest (0.8053) with no fine-tuning of neurons.
- **Impact of Interventions:** Statistical interventions generally degrade the model’s performance. The degradation becomes more significant with higher percentiles (e.g., P90 and P95), indicating that extreme activation values may negatively affect generalization.

- **Language-Specific Trends:** For *ur*, fine-tuning both English and Urdu neurons (*set6\_en + ur*) shows marginal improvement in baseline performance compared to other setups. However, statistical interventions still result in performance drops.
- **Fine-Tuning Trade-Offs:** Fine-tuning neurons specific to the target language (*set1\_vi*, *set6\_hi*, *set6\_ur*) offers better specialization for the target language. However, this comes at the cost of reduced generalization, as seen in the reduced accuracy for higher percentiles.

| Fine-tune Neurons | Eval Lang | No Int | Int by $\mu$ | Int by P75 | Int by P90 | Int by P95 |
|-------------------|-----------|--------|--------------|------------|------------|------------|
| null              | vi        | 0.8053 | 0.7955       | 0.7915     | 0.7903     | 0.7897     |
| set1_en           | vi        | 0.8017 | 0.7957       | 0.7867     | 0.7847     | 0.7847     |
| set1_vi           | vi        | 0.8007 | 0.7951       | 0.7881     | 0.7855     | 0.7845     |
| set1_en+vi        | vi        | 0.8009 | 0.7943       | 0.7865     | 0.7845     | 0.7833     |
| null              | hi        | 0.7501 | 0.7521       | 0.7496     | 0.7499     | 0.7506     |
| set1_en           | hi        | 0.7476 | 0.7456       | 0.7454     | 0.7448     | 0.7434     |
| set6_en           | hi        | 0.7491 | 0.7456       | 0.7464     | 0.7458     | 0.7441     |
| set6_hi           | hi        | 0.7492 | 0.7456       | 0.7461     | 0.7448     | 0.7431     |
| set6_en+hi        | hi        | 0.7488 | 0.7448       | 0.7451     | 0.7446     | 0.7431     |
| null              | ur        | 0.7002 | 0.7044       | 0.7028     | 0.6938     | 0.6904     |
| set6_en           | ur        | 0.6984 | 0.7041       | 0.7026     | 0.6964     | 0.6923     |
| set6_ur           | ur        | 0.6984 | 0.7054       | 0.7024     | 0.6952     | 0.6913     |
| set6_en+ur        | ur        | 0.6992 | 0.7061       | 0.7021     | 0.6954     | 0.6919     |

TABLE 5.2: XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English.

The results highlight the complexities of fine-tuning specific neurons for multilingual tasks. While targeted fine-tuning can improve certain aspects of language specialization, it often comes at the cost of generalization.

Table 5.3 extends the evaluation of fine-tuning configurations by including additional statistical interventions (P5, P10, P25, and P50) for Vietnamese (*vi*), Hindi (*hi*), and Urdu (*ur*). These interventions focus on lower percentile activation values, complementing the prior analysis of higher percentile interventions.

### Insights from Results of Intervention with Lower Quantiles

- **Low Percentile Interventions:** The impact of interventions at lower percentiles (P5, P10) varies significantly across languages. For instance, in *vi*, there is a notable drop in performance with P5 (0.7366) but a recovery at P25 (0.8055), suggesting that activation suppression may initially disrupt but later stabilize.

| Fine-tune Neurons | Eval Lang | No Int | Int by P5 | Int by P10 | Int by P25 | Int by P50 |
|-------------------|-----------|--------|-----------|------------|------------|------------|
| null              | vi        | 0.8053 | 0.7366    | 0.7773     | 0.8055     | 0.7971     |
| set1_en           | vi        | 0.8017 | 0.7432    | 0.7803     | 0.8027     | 0.7923     |
| set1_vi           | vi        | 0.8007 | 0.7426    | 0.7801     | 0.8029     | 0.7927     |
| set1_en+vi        | vi        | 0.8009 | 0.7426    | 0.7805     | 0.8027     | 0.7919     |
| null              | hi        | 0.7501 | 0.7488    | 0.7518     | 0.7496     | 0.7526     |
| set1_en           | hi        | 0.7476 | 0.7451    | 0.7468     | 0.7461     | 0.7451     |
| set6_en           | hi        | 0.7491 | 0.7464    | 0.7456     | 0.7476     | 0.7468     |
| set6_hi           | hi        | 0.7492 | 0.7461    | 0.7456     | 0.7464     | 0.7456     |
| set6_en+hi        | hi        | 0.7488 | 0.7456    | 0.7468     | 0.7462     | 0.7454     |
| null              | ur        | 0.7002 | 0.6861    | 0.6873     | 0.6971     | 0.7054     |
| set6_en           | ur        | 0.6984 | 0.6886    | 0.6904     | 0.7016     | 0.7031     |
| set6_ur           | ur        | 0.6984 | 0.6873    | 0.6913     | 0.7011     | 0.7034     |
| set6_en+ur        | ur        | 0.6992 | 0.6867    | 0.6893     | 0.7016     | 0.7042     |

TABLE 5.3: XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for LAPE method for Llama3.1. The training language is always English

- **Comparison Across Languages:** In *hi*, the effects of interventions are minimal, with performance remaining stable across percentiles. This indicates a higher resilience to neuron-level changes compared to *vi* and *ur*.
- **Urdu-Specific Trends:** In *ur*, P50 yields the best results (0.7054), highlighting that moderate percentile interventions may enhance generalization for certain low-resource languages.
- **Fine-Tuning Configurations:** Targeting both English and target language neurons (e.g., *set6\_en+ur*) continues to offer a balanced trade-off, with slight improvements in stability compared to single-language setups (e.g., *set6\_ur*).

### 5.3.3 Act Prob 90p Experiment Results using Lang Specific Neurons

Table 5.4 summarizes the results of the XNLI task using the `act_prob_90p` method for fine-tuning. The results evaluate the impact of interventions at various statistical activation thresholds (*Mean*, *P75*, *P90*, *P95*) for the evaluation languages *vi*, *hi*, and *ur*. Different fine-tuning configurations are compared, including null (no targeted neurons fine-tuned), English-specific neurons, target language-specific neurons, and both English and target language neurons combined.

These results highlight the nuanced impact of targeted fine-tuning on the XNLI task. While higher quantile interventions can improve robustness, they may also disrupt the generalization

capability of the model, as seen in *hi*. For *vi*, the results suggest that linguistic similarity to the training language (*en*) allows for better adaptation even with minimal fine-tuning. However, the relative stability of *ur* results indicates that certain languages may require alternative strategies, such as increasing data representation during training or exploring script-specific adaptations.

| Fine-tune Neurons | Eval Lang | No Int | Int by $\mu$ | Int by P75 | Int by P90 | Int by P95 |
|-------------------|-----------|--------|--------------|------------|------------|------------|
| null              | vi        | 0.8053 | 0.7821       | 0.7891     | 0.7931     | 0.7957     |
| en                | vi        | 0.8027 | 0.7801       | 0.7865     | 0.7907     | 0.7923     |
| vi                | vi        | 0.8005 | 0.7799       | 0.7857     | 0.7901     | 0.7907     |
| en+vi             | vi        | 0.8005 | 0.7799       | 0.7853     | 0.7897     | 0.7909     |
| null              | hi        | 0.7501 | 0.7411       | 0.7268     | 0.7186     | 0.7122     |
| en                | hi        | 0.7492 | 0.7386       | 0.7262     | 0.7214     | 0.7148     |
| hi                | hi        | 0.7486 | 0.7382       | 0.7262     | 0.7212     | 0.7148     |
| en+hi             | hi        | 0.7488 | 0.7372       | 0.7256     | 0.7214     | 0.7144     |
| null              | ur        | 0.7002 | 0.6971       | 0.6978     | 0.6932     | 0.6875     |
| en                | ur        | 0.6994 | 0.7006       | 0.6978     | 0.6937     | 0.6893     |
| ur                | ur        | 0.6988 | 0.7001       | 0.6982     | 0.6935     | 0.6925     |
| en+ur             | ur        | 0.6978 | 0.6992       | 0.6961     | 0.6927     | 0.6915     |

TABLE 5.4: XNLI task results (intervention by higher quantiles) after targeted neurons fine-tuning for act\_prob\_90p method for Llama3.1. The training language is always English

Table 5.5 summarizes the results of the XNLI task with interventions at lower quantiles (*P5*, *P10*, *P25*, *P50*). The results evaluate the effect of these interventions on the evaluation languages *vi*, *hi*, and *ur*. These configurations provide insights into how lower activation thresholds influence model performance.

### 5.3.4 Act Prob 90p Experiment Results using Task Specific Neurons

This section presents the experimental results of the Act Prob 90p method using task-specific neurons computed from the XNLI dataset. The experiments were conducted to evaluate the effect of targeted neuron interventions on fine-tuning performance. The evaluations were performed using Vietnamese (*vi*) as the target language. The results are summarized in Table 5.6.

Table 5.7 provides additional experimental results for the Act Prob 90p method using task-specific neurons with interventions applied at lower quantiles (P5 to P50). The evaluations were performed on the Vietnamese (*vi*) target language under different fine-tuning configurations.

| <b>Fine-tune Neurons</b> | <b>Eval Lang</b> | <b>Int by P5</b> | <b>Int by P10</b> | <b>Int by P25</b> | <b>Int by P50</b> |
|--------------------------|------------------|------------------|-------------------|-------------------|-------------------|
| null                     | vi               | 0.7653           | 0.7747            | 0.7801            | 0.7835            |
| en                       | vi               | 0.7667           | 0.7749            | 0.7789            | 0.7827            |
| vi                       | vi               | 0.7653           | 0.7743            | 0.7779            | 0.7817            |
| en+vi                    | vi               | 0.7663           | 0.7745            | 0.7795            | 0.7805            |
| null                     | hi               | 0.7501           | 0.7468            | 0.7422            | 0.7378            |
| en                       | hi               | 0.7481           | 0.7451            | 0.7426            | 0.7378            |
| hi                       | hi               | 0.7486           | 0.7456            | 0.7416            | 0.7374            |
| en+hi                    | hi               | 0.7491           | 0.7462            | 0.7428            | 0.7364            |
| null                     | ur               | 0.6948           | 0.6958            | 0.6954            | 0.6966            |
| en                       | ur               | 0.6948           | 0.6946            | 0.6956            | 0.7001            |
| ur                       | ur               | 0.6946           | 0.6952            | 0.6952            | 0.6998            |
| en+ur                    | ur               | 0.6942           | 0.6964            | 0.6948            | 0.6998            |

TABLE 5.5: XNLI task results (intervention by lower quantiles) after targeted neurons fine-tuning for act\_prob\_90p method for Llama3.1. The training language is always English

| <b>Fine-Tune XNLI Neurons</b> | <b>Eval Lang</b> | <b>No Int</b> | <b>Int by <math>\mu</math></b> | <b>Int by P75</b> | <b>Int by P90</b> | <b>Int by P95</b> |
|-------------------------------|------------------|---------------|--------------------------------|-------------------|-------------------|-------------------|
| null                          | vi               | 0.8053        | 0.7926                         | 0.7994            | 0.8014            | 0.8028            |
| en                            | vi               | 0.7956        | 0.7926                         | 0.8004            | 0.8006            | 0.8026            |
| vi                            | vi               | 0.7951        | 0.7926                         | 0.7998            | 0.7998            | 0.8014            |
| en+vi                         | vi               | 0.7921        | 0.7924                         | 0.7982            | 0.7921            | 0.8012            |

TABLE 5.6: XNLI task results (intervention by higher quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English

| <b>Fine-Tune XNLI Neurons</b> | <b>Eval Lang</b> | <b>Int by P5</b> | <b>Int by P10</b> | <b>Int by P25</b> | <b>Int by P50</b> |
|-------------------------------|------------------|------------------|-------------------|-------------------|-------------------|
| null                          | vi               | 0.7706           | 0.7801            | 0.7912            | 0.7936            |
| en                            | vi               | 0.7702           | 0.7791            | 0.7896            | 0.7926            |
| vi                            | vi               | 0.7688           | 0.7774            | 0.7896            | 0.7931            |

TABLE 5.7: XNLI task results (intervention by lower quantiles) for Act Prob 90p experiments using task-specific neurons for Llama3.1. The training language is always English

# Chapter 6

## Conclusions

### 6.1 Conclusion and Future Work

#### 6.1.1 Conclusion

This study explored various configurations and methodologies for fine-tuning and evaluating the Meta-Llama-3.1-8B model on the XNLI task using both language-specific and task-specific neurons. The experiments leveraged LoRA (Low-Rank Adaptation) to efficiently fine-tune specific layers and subsets of neurons, ensuring minimal computational overhead while retaining robust performance. Key findings from this work include:

- **Effectiveness of Statistical Interventions:** The use of statistical activation patterns, such as mean and higher quantiles (e.g., P75, P90, P95), did not demonstrate a significant consistent improvements across multiple configurations.
- **Language-Specific Neurons:** Results revealed that language-specific neurons computed using Wikipedia data provide strong baselines for zero-shot transfer to other languages. Fine-tuning targeted neurons, such as English-specific or Vietnamese-specific neurons, led to additional marginal improvements in performance.
- **Insights into Zero-Shot Transfer:** Unexpectedly, in some cases, the performance of zero-shot transfer models exceeded that of directly fine-tuned models. This behavior was notably observed in languages such as Urdu, suggesting the model’s intrinsic robustness in certain linguistic domains.
- **Intervention with Lower Quantiles:** Activating neurons based on lower quantiles (e.g., P5, P10) degraded the performance, but higher quantile interventions were found to be more effective for optimizing performance across the XNLI task.

### 6.1.2 Future Work

While the results achieved in this study are not promising, several paths for future exploration remain:

- **Attention Neuron Selection:** Future work could explore neuron selection at the attention layer.
- **Cross-Lingual Generalization:** Investigating the impact of fine-tuning on low-resource languages and analyzing how neuron interventions affect cross-lingual generalization remains an open challenge.
- **Integration with Larger Architectures:** Scaling the current methodologies to larger models (e.g., Meta-Llama-13B) or adapting them to more recent architectures could provide further insights into the scalability and generalization of the proposed approaches.
- **Fine-Tuning Granularity:** Exploring finer-grained levels of neuron manipulation, such as per-layer or per-head interventions, may yield better task alignment and performance across diverse benchmarks.
- **Interpretability and Explainability:** Further analysis is needed to understand the reasons behind certain surprising behaviors, such as why zero-shot transfer sometimes outperforms direct fine-tuning. Enhancing interpretability could shed light on the underlying mechanisms of neuron activations and their role in language understanding.

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